电动力学

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符号表

符号	含义
δ_{ij}	Kronecker 符号
$\cdot$	点乘
$\times$	叉乘
$\boldsymbol{a}$	矢量函数
f	标量函数
a_i	矢量 $\boldsymbol{a}$ 的第 i 个分量
δ_{ij}	克罗内克尔符号
$\delta(\boldsymbol{x})$	狄拉克函数
ε_{ijk}	Levi–Civita 符号 (三维)
ε_0	真空介电常数
x_i	第 i 个坐标分量
$\boldsymbol{e}$	单位矢量
∇	哈密顿算子
r	$\sqrt{x_i x_i}$
$\boldsymbol{r}$	$x_i \boldsymbol{e}_i$
$\boldsymbol{R}$	旋转矩阵
$\boldsymbol{\Phi}_E$	电场强度通量
ρ	电荷密度

1 矢量分析

1.1 定义

1.1.1 爱因斯坦求和约定

$$\sum_{i=1}^{3} a_i b_i \equiv a_i b_i \tag{1.1}$$

1.1.2 克罗内克尔符号

$$\delta_{ij} \equiv \begin{cases} 1 \ (i = j) \\ 0 \ (i \neq j) \end{cases} \tag{1.2}$$

性质:

$$a_i b_j \delta_{ij} = a_i b_i \tag{1.3}$$

$$\delta_{ij} \delta_{jk} = \delta_{ik} \tag{1.4}$$

1.1.3 点乘

$$\boldsymbol{a} \cdot \boldsymbol{b} \equiv a_i b_i \tag{1.5}$$

1.1.4 Levi-Civita 符号

$$\varepsilon_{ijk} \equiv \begin{pmatrix} \delta_{1i} & \delta_{2i} & \delta_{3i} \\ \delta_{1j} & \delta_{2j} & \delta_{3j} \\ \delta_{1k} & \delta_{2k} & \delta_{3k} \end{pmatrix} \tag{1.6}$$

性质:

$$\varepsilon_{ijk} = -\varepsilon_{ikj} = -\varepsilon_{kji} = -\varepsilon_{jik} \tag{1.7}$$

$$
\begin{aligned}
\varepsilon_{ijk}\varepsilon_{lmn} &= \begin{pmatrix} \delta_{1i} & \delta_{2i} & \delta_{3i} \\ \delta_{1j} & \delta_{2j} & \delta_{3j} \\ \delta_{1k} & \delta_{2k} & \delta_{3k} \end{pmatrix} \begin{pmatrix} \delta_{1i} & \delta_{2i} & \delta_{3i} \\ \delta_{1j} & \delta_{2j} & \delta_{3j} \\ \delta_{1k} & \delta_{2k} & \delta_{3k} \end{pmatrix} \\
&= \begin{pmatrix} \delta_{1i} & \delta_{2i} & \delta_{3i} \\ \delta_{1j} & \delta_{2j} & \delta_{3j} \\ \delta_{1k} & \delta_{2k} & \delta_{3k} \end{pmatrix}^T \begin{pmatrix} \delta_{1l} & \delta_{2l} & \delta_{3l} \\ \delta_{1m} & \delta_{2m} & \delta_{3m} \\ \delta_{1n} & \delta_{2n} & \delta_{3n} \end{pmatrix} \\
&= \begin{pmatrix} \delta_{1i} & \delta_{1j} & \delta_{1k} \\ \delta_{2i} & \delta_{2j} & \delta_{2k} \\ \delta_{3i} & \delta_{3j} & \delta_{3k} \end{pmatrix} \begin{pmatrix} \delta_{1l} & \delta_{2l} & \delta_{3l} \\ \delta_{1m} & \delta_{2m} & \delta_{3m} \\ \delta_{1n} & \delta_{2n} & \delta_{3n} \end{pmatrix} \\
&= \begin{pmatrix} \delta_{1l} & \delta_{2l} & \delta_{3l} \\ \delta_{1m} & \delta_{2m} & \delta_{3m} \\ \delta_{1n} & \delta_{2n} & \delta_{3n} \end{pmatrix} \begin{pmatrix} \delta_{1i} & \delta_{1j} & \delta_{1k} \\ \delta_{2i} & \delta_{2j} & \delta_{2k} \\ \delta_{3i} & \delta_{3j} & \delta_{3k} \end{pmatrix} \\
&= \begin{pmatrix} \delta_{li} & \delta_{lj} & \delta_{lk} \\ \delta_{mi} & \delta_{mj} & \delta_{mk} \\ \delta_{ni} & \delta_{nj} & \delta_{nk} \end{pmatrix}
\end{aligned} \tag{1.8}
$$

$$
\varepsilon_{ijk}\varepsilon_{lmk} = \sum_{k=1}^{3} \begin{pmatrix} \delta_{li} & \delta_{lj} & \delta_{lk} \\ \delta_{mi} & \delta_{mj} & \delta_{mk} \\ \delta_{ki} & \delta_{kj} & 1 \end{pmatrix} = \begin{pmatrix} \delta_{li} & \delta_{lj} & 0 \\ \delta_{mi} & \delta_{mj} & 0 \\ 0 & 0 & 1 \end{pmatrix} = \delta_{li}\delta_{mj} - \delta_{lj}\delta_{mi} \tag{1.9}
$$

$$
\varepsilon_{ijk}\varepsilon_{ljk} = \delta_{li}\delta_{jj} - \delta_{lj}\delta_{ji} = 3\delta_{li} - \delta_{li} = 2\delta_{li} \tag{1.10}
$$

1.1.5 叉乘

$$
\boldsymbol{a} \times \boldsymbol{b} \equiv \varepsilon_{ijk} a_i b_j \boldsymbol{e}_k = \begin{pmatrix} \boldsymbol{e}_i & \boldsymbol{e}_j & \boldsymbol{e}_k \\ a_i & a_j & a_k \\ b_i & b_j & b_k \end{pmatrix} \tag{1.11}
$$

性质:

$$
\begin{aligned}
\boldsymbol{c}\cdot(\boldsymbol{a}\times\boldsymbol{b}) &= \boldsymbol{c}\cdot\varepsilon_{ijk}a_ib_j\boldsymbol{e}_k \\
&= c_k\varepsilon_{ijk}a_ib_j \\
&= \varepsilon_{ijk}a_ib_jc_k \\
&= \begin{pmatrix} \delta_{1i} & \delta_{2i} & \delta_{3i} \\ \delta_{1j} & \delta_{2j} & \delta_{3j} \\ \delta_{1k} & \delta_{2k} & \delta_{3k} \end{pmatrix} a_ib_jc_k \\
&= \begin{pmatrix} \delta_{1i}a_i & \delta_{2i}a_i & \delta_{3i}a_i \\ \delta_{1j}b_j & \delta_{2j}b_j & \delta_{3j}b_j \\ \delta_{1k}c_k & \delta_{2k}c_k & \delta_{3k}c_k \end{pmatrix} \\
&= \begin{pmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{pmatrix}
\end{aligned} \tag{1.12}
$$

$$
\begin{aligned}
\boldsymbol{c}\times(\boldsymbol{a}\times\boldsymbol{b}) &= \boldsymbol{c}\times\varepsilon_{ijk}a_ib_j\boldsymbol{e}_k \\
&= \varepsilon_{lkn}c_l\varepsilon_{ijk}a_ib_j\boldsymbol{e}_n \\
&= -\varepsilon_{lnk}\varepsilon_{ijk}a_ib_jc_l\boldsymbol{e}_n \\
&= -(\delta_{il}\delta_{jn}-\delta_{in}\delta_{jl})a_ib_jc_l\boldsymbol{e}_n \\
&= -\delta_{il}\delta_{jn}a_ib_jc_l\boldsymbol{e}_n+\delta_{in}\delta_{jl}a_ib_jc_l\boldsymbol{e}_n \\
&= -a_ib_jc_i\boldsymbol{e}_j+a_ib_jc_j\boldsymbol{e}_i \\
&= -(\boldsymbol{a}\cdot\boldsymbol{c})\boldsymbol{b}+(\boldsymbol{b}\cdot\boldsymbol{c})\boldsymbol{a}
\end{aligned} \tag{1.13}
$$

$$
\boldsymbol{e}\times(\boldsymbol{e}\times\boldsymbol{b}) = -(\boldsymbol{e}\cdot\boldsymbol{e})\boldsymbol{b}+(\boldsymbol{b}\cdot\boldsymbol{e})\boldsymbol{e} = -\boldsymbol{b}+(\boldsymbol{b}\cdot\boldsymbol{e})\boldsymbol{e} \tag{1.14}
$$

1.1.6 哈密顿算子

$$
\boldsymbol{\nabla} \equiv \frac{\partial}{\partial x_i}\boldsymbol{e}_i \tag{1.15}
$$

1.1.7 梯度

设位置矢量 $\boldsymbol{r}=\boldsymbol{r}(u_1,u_2,u_3)$, 拉梅系数为

$$
h_i \equiv \left\|\frac{\partial\boldsymbol{r}}{\partial u_i}\right\|, \qquad \boldsymbol{e} \equiv \frac{1}{h_i}\frac{\partial\boldsymbol{r}}{\partial u_i}
$$

且 $\boldsymbol{e}_i\cdot\boldsymbol{e}_j=\delta_{ij}$。

对标量场 $f(u_1,u_2,u_3)$, 全微分为

$$
\mathrm{d}f = \frac{\partial f}{\partial u_i}\,\mathrm{d}u_i.
$$

无穷小位移 (线元) 为

$$\mathrm{d}\boldsymbol{l} = h_i\ \mathrm{d}u_i\,\boldsymbol{e}_i.$$

设 $\boldsymbol{\nabla} f = G_i\boldsymbol{e}_i$, 则

$$\mathrm{d}f = \boldsymbol{\nabla} f\cdot\boldsymbol{dl} = G_i h_i\ \mathrm{d}u_i.$$

比较系数, 得 $\boldsymbol{G} = \dfrac{1}{h_i}\dfrac{\partial f}{\partial u_i}\boldsymbol{e}_i$, 从而

$$\boldsymbol{\nabla} f = \frac{1}{h_i}\frac{\partial f}{\partial u_i}\boldsymbol{e}_i \tag{1.16}$$

对于直角坐标

$$\boldsymbol{\nabla} f = \frac{\partial f}{\partial x_i}\boldsymbol{e}_i \tag{1.17}$$

性质:

$$\boldsymbol{\nabla} r = \frac{\partial\sqrt{x_i x_i}}{\partial x_i}\boldsymbol{e}_i = \frac{x_i}{r}\boldsymbol{e}_i = \frac{\boldsymbol{r}}{r} \tag{1.18}$$

$$\boldsymbol{\nabla} f(r) = \frac{\partial f}{\partial r}\frac{\partial r}{\partial x_i}\boldsymbol{e}_i = \frac{\partial f}{\partial r}\frac{\boldsymbol{r}}{r} \tag{1.19}$$

1.1.8 散度

对微小增量 $\mathrm{d}u_1, \mathrm{d}u_2, \mathrm{d}u_3$, 线元在三方向的实际长度分别为

$$\mathrm{d}l_1 = h_1\,\mathrm{d}u_1,\quad \mathrm{d}l_2 = h_2\,\mathrm{d}u_2,\quad \mathrm{d}l_3 = h_3\,\mathrm{d}u_3$$

因此微小长方体的体元为

$$\mathrm{d}\tau = \mathrm{d}l_1\,\mathrm{d}l_2\,\mathrm{d}l_3 = h_1h_2h_3\,\mathrm{d}u_1\,\mathrm{d}u_2\,\mathrm{d}u_3. \tag{1.20}$$

设矢量场在该点为

$$\boldsymbol{a} = a_1\boldsymbol{e}_1 + a_2\boldsymbol{e}_2 + a_3\boldsymbol{e}_3,$$

其中 a_1, a_2, a_3 为在局部单位基矢下的分量 (均为 u_1, u_2, u_3 的函数)。

下面计算穿过微小长方体六个面的净通量。以 u_1 方向为例, 考察在 u_1 与 $u_1 + \mathrm{d}u_1$ 两个面:

面 u_1(在坐标 u 处, 向外法向为 $-\boldsymbol{e}_1$)：面积元为

$$\mathrm{d}\boldsymbol{a}_1^{(-)} = -\boldsymbol{e}_1(h_2\,\mathrm{d}u_2)(h_3\,\mathrm{d}u_3) = -\boldsymbol{e}_1h_2h_3\,\mathrm{d}u_2\,\mathrm{d}u_3,$$

对应的通量 (近似取该面中心处分量)：

$$\Phi_1^{(-)} = \boldsymbol{a}(u_1,u_2,u_3)\cdot\mathrm{d}\boldsymbol{a}_1^{(-)} = -a_1(u_1,u_2,u_3)h_2h_3\,\mathrm{d}u_2\,\mathrm{d}u_3.$$

面 $u + \mathrm{d}u$(向外法向为 $+\boldsymbol{e}_1$)：面积元

$$\mathrm{d}\boldsymbol{a}_1^{(+)} = +\boldsymbol{e}_1h_2h_3\,\mathrm{d}u_2\,\mathrm{d}u_3,$$

通量 (在 $u + du$ 处)：

$$\Phi_1^{(+)} = a_1(u_1+du_1,u_2,u_3)h_2(u_1+du_1,u_2,u_3)h_3(u_1+du_1,u_2,u_3)\,\mathrm{d}u_2\,\mathrm{d}u_3.$$

因此穿过这对面的净通量为

$$\begin{aligned}\Delta\Phi_1 &= \Phi_1^{(+)} - \Phi_1^{(-)}\\ &= [a_1(u_1+\mathrm{d}u_1,u_2,u_3)h_2(u_1+\mathrm{d}u_1,u_2,u_3)h_3(u_1+\mathrm{d}u_1,u_2,u_3) - a_1(u_1,u_2,u_3)h_2h_3]\,\mathrm{d}u_2\,\mathrm{d}u_3.\end{aligned}$$

用泰勒展开到一阶并忽略高阶项, 得

$$\Delta\Phi_u = \frac{\partial}{\partial u_1}(h_2h_3a_1)\,\mathrm{d}u_1\,\mathrm{d}u_2\,\mathrm{d}u_3 + o(\mathrm{d}u_1\,\mathrm{d}u_2\,\mathrm{d}u_3).$$

对 u_2 和 u_3 方向做同样的计算, 六个面的净通量近似为三项之和：

$$\Delta\Phi_{\text{total}} = \frac{\partial}{\partial u_i}\left(h_1h_2h_3\frac{a_i}{h_i}\right)\mathrm{d}u_1\,\mathrm{d}u_2\,\mathrm{d}u_3$$

由于体元按照式(1.20)为 $\mathrm{d}\tau = h_1h_2h_3\,\mathrm{d}u_1\,\mathrm{d}u_2\,\mathrm{d}u_3$, 故单位体积的通量密度 (即散度) 为极限：

$$\begin{aligned}\boldsymbol{\nabla}\cdot\boldsymbol{a} &= \lim_{\Delta\tau\to 0}\frac{\Delta\Phi_{\text{total}}}{d\tau}\\ &= \frac{1}{h_1h_2h_3}\frac{\partial}{\partial u_i}\left(h_1h_2h_3\frac{a_i}{h_i}\right).\end{aligned}$$

于是得到正交曲线坐标系下的散度公式：

$$\boldsymbol{\nabla}\cdot\boldsymbol{a} = \frac{1}{h_1h_2h_3}\frac{\partial}{\partial u_i}\left(h_1h_2h_3\frac{a_i}{h_i}\right). \tag{1.21}$$

对于直角坐标

$$\boldsymbol{\nabla}\cdot\boldsymbol{a} = \frac{\partial a_i}{\partial x_i} \tag{1.22}$$

性质:

$$\boldsymbol{\nabla}\cdot\boldsymbol{r} = 3 \tag{1.23}$$

$$\boldsymbol{\nabla}\cdot\boldsymbol{f}(r) = \frac{\partial f_i(r)}{\partial x_i} = f_i'\frac{\partial r}{\partial x_i} = \boldsymbol{f}'\cdot\frac{\boldsymbol{r}}{r} \tag{1.24}$$

1.1.9 旋度

选取微小矩形位于 u_1-u_2 平面, 固定 u_3。矩形四边的线元素：

$$\begin{aligned}
&\text{下边: } \mathrm{d}\boldsymbol{l}_1 = h_1\,\mathrm{d}u_1\boldsymbol{e}_1, \quad \text{从}(u_1,u_2)\to(u_1+\mathrm{d}u_1,u_2)\\
&\text{右边: } \mathrm{d}\boldsymbol{l}_2 = h_2\,\mathrm{d}u_2\boldsymbol{e}_2, \quad \text{从}(u_1+\mathrm{d}u_1,u_2)\to(u_1+\mathrm{d}u_1,u_2+\mathrm{d}u_2)\\
&\text{上边: } \mathrm{d}\boldsymbol{l}_3 = -h_1\,\mathrm{d}u_1\boldsymbol{e}_1, \quad \text{从}(u_1+\mathrm{d}u_1,u_2+\mathrm{d}u_2)\to(u_1,u_2+\mathrm{d}u_2)\\
&\text{左边: } \mathrm{d}\boldsymbol{l}_4 = -h_2du_2\boldsymbol{e}_2, \quad \text{从}(u_1,u_2+\mathrm{d}u_2)\to(u_1,u_2)
\end{aligned}$$

沿每一边计算 $\boldsymbol{a}\cdot d\boldsymbol{l}$ 并做泰勒展开 (只保留一阶项)：

$$\oint\boldsymbol{a}\cdot\mathrm{d}\boldsymbol{l} \approx \left[\frac{\partial}{\partial u_1}(h_2a_2) - \frac{\partial}{\partial u_2}(h_1a_1)\right]\mathrm{d}u_1\,\mathrm{d}u_2$$

面元面积：

$$dS_3 = h_1h_2\,\mathrm{d}u_1\,\mathrm{d}u_2$$

旋度在 $\boldsymbol{e}_3$ 方向的分量：

$$(\boldsymbol{\nabla}\times\boldsymbol{a})_3 = \lim_{\mathrm{d}u_1,\mathrm{d}u_2\to 0}\frac{\oint\boldsymbol{a}\cdot\mathrm{d}\boldsymbol{l}}{\mathrm{d}S_3} = \frac{1}{h_1h_2}\left[\frac{\partial}{\partial u_1}(h_2a_2) - \frac{\partial}{\partial u_2}(h_1a_1)\right]$$

同理, 对于其他两个方向：

$$(\boldsymbol{\nabla}\times\boldsymbol{a})_1 = \frac{1}{h_2h_3}\left[\frac{\partial}{\partial u_2}(h_3a_3) - \frac{\partial}{\partial u_3}(h_2a_2)\right], \quad (\boldsymbol{\nabla}\times\boldsymbol{a})_2 = \frac{1}{h_3h_1}\left[\frac{\partial}{\partial u_3}(h_1a_1) - \frac{\partial}{\partial u_1}(h_3a_3)\right]$$

最终正交曲线坐标系下旋度公式为：

$$\boldsymbol{\nabla} \times \boldsymbol{a} = \frac{h_i}{h_1 h_2 h_3} \varepsilon_{ijk} \frac{\partial (h_k a_k)}{\partial u_j} \boldsymbol{e}_i \tag{1.25}$$

对于直角坐标

$$\boldsymbol{\nabla} \times \boldsymbol{a} = \varepsilon_{ijk} \frac{\partial a_j}{\partial x_i} \boldsymbol{e}_k \tag{1.26}$$

性质:

$$\boldsymbol{\nabla} \times \boldsymbol{r} = 0 \tag{1.27}$$

$$\boldsymbol{\nabla} \times (f\boldsymbol{a}) = \varepsilon_{ijk} \frac{\partial (f a_j)}{\partial x_i} \boldsymbol{e}_k = f \varepsilon_{ijk} \frac{\partial a_j}{\partial x_i} \boldsymbol{e}_k + \varepsilon_{ijk} \frac{\partial f}{\partial x_i} a_j \boldsymbol{e}_k = f(\boldsymbol{\nabla} \times \boldsymbol{a}) + (\boldsymbol{\nabla} f) \times \boldsymbol{a} \tag{1.28}$$

1.1.10 并积

$$\boldsymbol{\nabla} \boldsymbol{a} \equiv \begin{pmatrix} \frac{\partial a_1}{x_1} & \frac{\partial a_2}{x_1} & \frac{\partial a_3}{x_1} \\ \frac{\partial a_1}{x_2} & \frac{\partial a_2}{x_2} & \frac{\partial a_3}{x_2} \\ \frac{\partial a_1}{x_3} & \frac{\partial a_2}{x_3} & \frac{\partial a_3}{x_3} \end{pmatrix} = \frac{\partial a_i}{\partial x_j} \boldsymbol{e}_i \boldsymbol{e}_j \tag{1.29}$$

1.1.11 拉普拉斯算子

定义为梯度的散度, 由1.16和1.21得

$$\begin{aligned} \boldsymbol{\nabla}^2 f &= \boldsymbol{\nabla} \cdot (\boldsymbol{\nabla} f) \\ &= \boldsymbol{\nabla} \cdot \left(\frac{1}{h_i} \frac{\partial f}{\partial u_i} \right) \\ &= \frac{1}{h_1 h_2 h_3} \frac{\partial}{\partial u_i} \left(h_1 h_2 h_3 \frac{\frac{1}{h_i} \frac{\partial f}{\partial u_i}}{h_i} \right) \\ &= \frac{1}{h_1 h_2 h_3} \frac{\partial}{\partial u_i} \left(h_1 h_2 h_3 \frac{\partial f}{h_i^2 \partial u_i} \right) \end{aligned} \tag{1.30}$$

对于直角坐标

$$\boldsymbol{\nabla}^2 = \frac{\partial^2}{\partial x_i^2} \tag{1.31}$$

性质:

$$
\begin{aligned}
\boldsymbol{\nabla}^2(fg) &= \frac{\partial}{\partial x_i}\frac{\partial fg}{\partial x_i} \\
&= \frac{\partial}{\partial x_i}\left(g\frac{\partial f}{\partial x_i} + f\frac{\partial g}{\partial x_i}\right) \\
&= \frac{\partial}{\partial x_i}\left(g\frac{\partial f}{\partial x_i}\right) + \frac{\partial}{\partial x_i}\left(f\frac{\partial g}{\partial x_i}\right) \\
&= \frac{\partial}{\partial x_i}\left(g\frac{\partial f}{\partial x_i}\right) + \frac{\partial}{\partial x_i}\left(f\frac{\partial g}{\partial x_i}\right) \\
&= \frac{\partial g}{\partial x_i}\frac{\partial f}{\partial x_i} + g\frac{\partial^2 f}{\partial x_i^2} + \frac{\partial f}{\partial x_i}\frac{\partial g}{\partial x_i} + a\frac{\partial^2 g}{\partial x_i^2} \\
&= g\frac{\partial^2 f}{\partial x_i^2} + 2\frac{\partial f}{\partial x_i}\frac{\partial g}{\partial x_i} + f\frac{\partial^2 g}{\partial x_i^2} \\
&= g\boldsymbol{\nabla}^2 f + 2(\boldsymbol{\nabla} f)\cdot(\boldsymbol{\nabla} g) + f\boldsymbol{\nabla}^2 g
\end{aligned} \tag{1.32}
$$

1.2 微分运算

1.2.1 和规则

$$\boldsymbol{\nabla}(f+g) = \boldsymbol{\nabla} f + \boldsymbol{\nabla} g \tag{1.33}$$

$$\boldsymbol{\nabla}\cdot(\boldsymbol{a}+\boldsymbol{b}) = \boldsymbol{\nabla}\cdot\boldsymbol{a} + \boldsymbol{\nabla}\cdot\boldsymbol{b} \tag{1.34}$$

$$\boldsymbol{\nabla}\times(\boldsymbol{a}+\boldsymbol{b}) = \boldsymbol{\nabla}\times\boldsymbol{a} + \boldsymbol{\nabla}\times\boldsymbol{b} \tag{1.35}$$

1.2.2 积规则

$$\boldsymbol{\nabla}(fg) = g\boldsymbol{\nabla} f + f\boldsymbol{\nabla} g \tag{1.36}$$

$$
\begin{aligned}
&\boldsymbol{a}\times(\boldsymbol{\nabla}\times\boldsymbol{b}) + \boldsymbol{b}\times(\boldsymbol{\nabla}\times\boldsymbol{a}) + (\boldsymbol{a}\cdot\boldsymbol{\nabla})\boldsymbol{b} + (\boldsymbol{b}\cdot\boldsymbol{\nabla})\boldsymbol{a} \\
=&\boldsymbol{a}\times\left(\varepsilon_{ijk}\frac{\partial b_j}{\partial x_i}\boldsymbol{e}_k\right) + \boldsymbol{b}\times\left(\varepsilon_{ijk}\frac{\partial a_j}{\partial x_i}\boldsymbol{e}_k\right) + \left(a_i\frac{\partial}{\partial x_i}\right)\boldsymbol{b} + \left(b_i\frac{\partial}{\partial x_i}\right)\boldsymbol{a} \\
=&\varepsilon_{lkn}a_l\varepsilon_{ijk}\frac{\partial b_j}{\partial x_i}\boldsymbol{e}_n + \varepsilon_{lkn}b_l\varepsilon_{ijk}\frac{\partial a_j}{\partial x_i}\boldsymbol{e}_n + a_i\frac{\partial b_j}{\partial x_i}\boldsymbol{e}_j + b_i\frac{\partial a_j}{\partial x_i}\boldsymbol{e}_j \\
=&-\varepsilon_{lnk}\varepsilon_{ijk}a_l\frac{\partial b_j}{\partial x_i}\boldsymbol{e}_n - \varepsilon_{lnk}\varepsilon_{ijk}b_l\frac{\partial a_j}{\partial x_i}\boldsymbol{e}_n + a_i\frac{\partial b_j}{\partial x_i}\boldsymbol{e}_j + b_i\frac{\partial a_j}{\partial x_i}\boldsymbol{e}_j \\
=&-(\delta_{il}\delta_{jn} - \delta_{in}\delta_{jl})a_l\frac{\partial b_j}{\partial x_i}\boldsymbol{e}_n - (\delta_{il}\delta_{jn} - \delta_{in}\delta_{jl})b_l\frac{\partial a_j}{\partial x_i}\boldsymbol{e}_n + a_i\frac{\partial b_j}{\partial x_i}\boldsymbol{e}_j + b_i\frac{\partial a_j}{\partial x_i}\boldsymbol{e}_j \\
=&\left(-\delta_{il}\delta_{jn}a_l\frac{\partial b_j}{\partial x_i}\boldsymbol{e}_n + \delta_{in}\delta_{jl}a_l\frac{\partial b_j}{\partial x_i}\boldsymbol{e}_n\right) + \left(-\delta_{il}\delta_{jn}b_l\frac{\partial a_j}{\partial x_i}\boldsymbol{e}_n + \delta_{in}\delta_{jl}b_l\frac{\partial a_j}{\partial x_i}\boldsymbol{e}_n\right) + a_i\frac{\partial b_j}{\partial x_i}\boldsymbol{e}_j + b_i\frac{\partial a_j}{\partial x_i}\boldsymbol{e}_j \\
=&\left(-a_i\frac{\partial b_j}{\partial x_i}\boldsymbol{e}_j + a_j\frac{\partial b_j}{\partial x_i}\boldsymbol{e}_i\right) + \left(-b_i\frac{\partial a_j}{\partial x_i}\boldsymbol{e}_j + b_j\frac{\partial a_j}{\partial x_i}\boldsymbol{e}_i\right) + a_i\frac{\partial b_j}{\partial x_i}\boldsymbol{e}_j + b_i\frac{\partial a_j}{\partial x_i}\boldsymbol{e}_j \\
=&a_j\frac{\partial b_j}{\partial x_i}\boldsymbol{e}_i + b_j\frac{\partial a_j}{\partial x_i}\boldsymbol{e}_i \\
=&\frac{\partial(a_j b_j)}{\partial x_i}\boldsymbol{e}_i \\
=&\boldsymbol{\nabla}(\boldsymbol{a}\cdot\boldsymbol{b})
\end{aligned} \tag{1.37}
$$

$$\boldsymbol{\nabla}\cdot(f\boldsymbol{a})=\frac{\partial f a_i}{\partial x_i}=f\frac{\partial a_i}{\partial x_i}+a_i\frac{\partial f}{\partial x_i}=f\boldsymbol{\nabla}\cdot\boldsymbol{a}+\boldsymbol{a}\cdot\boldsymbol{\nabla}f \tag{1.38}$$

$$\begin{aligned}\boldsymbol{b}\cdot(\boldsymbol{\nabla}\times\boldsymbol{a})-\boldsymbol{a}\cdot(\boldsymbol{\nabla}\times\boldsymbol{b})&=\boldsymbol{b}\cdot(\varepsilon_{ijk}\frac{\partial a_j}{\partial x_i}\boldsymbol{e}_k)-\boldsymbol{a}\cdot(\varepsilon_{ijk}\frac{\partial b_j}{\partial x_i}\boldsymbol{e}_k)\\&=b_k\varepsilon_{ijk}\frac{\partial a_j}{\partial x_i}-a_k\varepsilon_{ijk}\frac{\partial b_j}{\partial x_i}\\&=b_j\varepsilon_{ijk}\frac{\partial a_i}{\partial x_k}+a_i\varepsilon_{ijk}\frac{\partial b_j}{\partial x_k}\\&=\frac{\partial\varepsilon_{ijk}a_ib_j}{\partial x_k}\\&=\boldsymbol{\nabla}\cdot(\varepsilon_{ijk}a_ib_j\boldsymbol{e}_k)\\&=\boldsymbol{\nabla}\cdot(\boldsymbol{a}\times\boldsymbol{b})\end{aligned} \tag{1.39}$$

1.2.3 二阶积规则

$$\boldsymbol{\nabla}\times(\boldsymbol{\nabla}f)=\boldsymbol{\nabla}\times(\frac{\partial f}{\partial x_i}\boldsymbol{e}_i)=\varepsilon_{ijk}\frac{\partial}{\partial x_i}\frac{\partial f}{\partial x_j}\boldsymbol{e}_k=0 \tag{1.40}$$

$$\boldsymbol{\nabla}\cdot(\boldsymbol{\nabla}\times\boldsymbol{a})=\boldsymbol{\nabla}\cdot(\varepsilon_{ijk}\frac{\partial a_i}{\partial x_j}\boldsymbol{e}_k)=\varepsilon_{ijk}\frac{\partial^2 a_i}{\partial x_j\partial x_k}=0 \tag{1.41}$$

$$\Delta f=\frac{\partial^2 f}{\partial x_i^2} \tag{1.42}$$

$$\Delta\boldsymbol{a}=\frac{\partial^2(a_j\boldsymbol{e}_j)}{\partial x_i^2} \tag{1.43}$$

$$\begin{aligned}\boldsymbol{\nabla}\times(\boldsymbol{\nabla}\times\boldsymbol{a})&=\boldsymbol{\nabla}\times(\varepsilon_{ijk}\frac{\partial a_i}{\partial x_j}\boldsymbol{e}_k)\\&=\varepsilon_{lmn}\varepsilon_{ijl}\frac{\partial^2 a_i}{\partial x_j\partial x_m}\boldsymbol{e}_n\\&=\varepsilon_{mnl}\varepsilon_{ijl}\frac{\partial^2 a_i}{\partial x_j\partial x_m}\boldsymbol{e}_n\\&=(\delta_{im}\delta_{jn}-\delta_{in}\delta_{jm})\frac{\partial^2 a_i}{\partial x_j\partial x_m}\boldsymbol{e}_n\\&=\delta_{im}\delta_{jn}\frac{\partial^2 a_i}{\partial x_j\partial x_m}\boldsymbol{e}_n-\delta_{in}\delta_{jm}\frac{\partial^2 a_i}{\partial x_j\partial x_m}\boldsymbol{e}_n\\&=\frac{\partial^2 a_i}{\partial x_j\partial x_i}\boldsymbol{e}_j-\frac{\partial^2 a_i}{\partial x_j\partial x_j}\boldsymbol{e}_i\\&=\boldsymbol{\nabla}(\boldsymbol{\nabla}\cdot\boldsymbol{a})-\Delta\boldsymbol{a}\end{aligned} \tag{1.44}$$

1.3 积分运算

1.3.1 高斯定理

我们已经得到正交坐标系中的散度局部公式1.21。因此, 对体积元 $\Delta\tau=h_1h_2h_3\,\Delta u_1\Delta u_2\Delta u_3$ 的六个面计算净通量时：

$$\Phi^{(\mathrm{out})}-\Phi^{(\mathrm{in})}=\left[\frac{\partial}{\partial u_i}\left(h_1h_2h_3\frac{a_i}{h_i}\right)\right]\Delta u_1\Delta u_2\Delta u_3=(\boldsymbol{\nabla}\cdot\boldsymbol{a})\,\Delta\tau.$$

将整个区域 V 分割成许多此类体积元, 内部公共面的通量相互抵消, 仅留下外边界 ∂V 的通量。令分割尺寸趋于零, 通量和体积和分别趋于各自积分, 故得到

$$\iiint_V (\boldsymbol{\nabla}\cdot\boldsymbol{a})\,\mathrm{d}V = \iint_{\partial V} \boldsymbol{a}\cdot\mathrm{d}\boldsymbol{S} \tag{1.45}$$

1.3.2 斯托克斯定理

我们已经得到任意正交坐标系中的旋度局部公式1.25 , 考虑由 (u_1, u_2) 增量生成的一个无穷小曲面元

$$\mathrm{d}\boldsymbol{S} = (h_1 h_2\,\mathrm{d}u_1\,\mathrm{d}u_2)\boldsymbol{e}_3.$$

沿此小曲面元的边界进行线积分, 可得

$$\oint\!\!\!\oint_{\partial S} \boldsymbol{a}\cdot d\boldsymbol{l} = (\nabla\times\boldsymbol{a})\cdot d\boldsymbol{S}.$$

将整个曲面分割成许多这样的小元, 内部公共边界的线积分相互抵消, 仅留下外边界 ∂S 的线积分。令分割尺寸趋于零, 总和趋于积分, 故得

$$\iint_S (\boldsymbol{\nabla}\times\boldsymbol{a})\cdot\mathrm{d}\boldsymbol{S} = \oint\!\!\!\oint_{\partial S} \boldsymbol{a}\cdot\mathrm{d}\boldsymbol{l} \tag{1.46}$$

1.4 曲面坐标系

1.4.1 柱坐标系

将柱坐标系坐标转化为直角系坐标的方式为

$$x_1 = s\cos\phi,\ x_2 = s\sin\phi,\ x_3 = x_3 \tag{1.47}$$

其中 s 是坐标点到 z 轴的距离。柱坐标系和直角系坐标无限小元转换方式为

$$\begin{pmatrix}\mathrm{d}x_1\\ \mathrm{d}x_2\\ \mathrm{d}x_3\end{pmatrix} = \begin{pmatrix}\frac{x_1}{s} & -x_2 & 0\\ \frac{x_2}{s} & x_1 & 0\\ 0 & 0 & 1\end{pmatrix}\begin{pmatrix}\mathrm{d}s\\ \mathrm{d}\phi\\ \mathrm{d}x_3\end{pmatrix} \tag{1.48}$$

其拉梅系数为

$$h_1 = 1,\ h_2 = s,\ h_3 = 1 \tag{1.49}$$

其无限小位移为

$$\mathrm{d}\boldsymbol{l} = h_i\,\mathrm{d}u_i\boldsymbol{e}_i = \mathrm{d}s\boldsymbol{s} + s\,\mathrm{d}\phi\boldsymbol{\phi} + \mathrm{d}x_3\boldsymbol{x}_3 \tag{1.50}$$

其体积元为

$$\mathrm{d}V = s\,\mathrm{d}s\,\mathrm{d}\phi\,\mathrm{d}x_3 \tag{1.51}$$

梯度

$$\nabla f = \frac{1}{h_i}\frac{\partial f}{\partial u_i}\boldsymbol{e}_i = \frac{\partial}{\partial s}\boldsymbol{s} + \frac{1}{s}\frac{\partial}{\partial\phi}\boldsymbol{\phi} + \frac{\partial}{\partial x_3}\boldsymbol{x}_3 \tag{1.52}$$

散度

$$\boldsymbol{\nabla}\cdot\boldsymbol{a} = \frac{1}{s}(\frac{\partial s a_s}{\partial s} + \frac{\partial a_\phi}{\partial\phi} + \frac{\partial s a_3}{\partial x_3}) = \frac{1}{s}\frac{\partial s a_s}{\partial s} + \frac{1}{s}\frac{\partial a_\phi}{\partial\phi} + \frac{\partial a_3}{\partial x_3} = \frac{a_s}{s} + \frac{\partial a_s}{\partial s} + \frac{1}{s}\frac{\partial a_\phi}{\partial\phi} + \frac{\partial a_3}{\partial x_3} \tag{1.53}$$

旋度

$$
\begin{aligned}
\boldsymbol{\nabla}\times\boldsymbol{a} &= \frac{h_i}{h_1h_2h_3}\varepsilon_{ijk}\frac{\partial(h_ka_k)}{\partial u_j}\boldsymbol{e}_i \\
&= \frac{h_i}{s}\varepsilon_{ijk}\frac{\partial(h_ka_k)}{\partial u_j}\boldsymbol{e}_i \\
&= \frac{1}{s}\left(\varepsilon_{123}\frac{\partial a_3}{\partial u_2}\boldsymbol{e}_1+\varepsilon_{132}\frac{\partial sa_2}{\partial u_3}\boldsymbol{e}_1+\varepsilon_{321}\frac{\partial a_1}{\partial u_2}\boldsymbol{e}_3+s\varepsilon_{213}\frac{\partial a_3}{\partial u_1}\boldsymbol{e}_2+\varepsilon_{312}\frac{\partial sa_2}{\partial u_1}\boldsymbol{e}_3+s\varepsilon_{231}\frac{\partial a_1}{\partial u_3}\boldsymbol{e}_2\right) \\
&= \frac{1}{s}\left(\frac{\partial a_3}{\partial u_2}\boldsymbol{e}_1-\frac{\partial sa_2}{\partial u_3}\boldsymbol{e}_1-\frac{\partial a_1}{\partial u_2}\boldsymbol{e}_3-s\frac{\partial a_3}{\partial u_1}\boldsymbol{e}_2+\frac{\partial sa_2}{\partial u_1}\boldsymbol{e}_3+s\frac{\partial a_1}{\partial u_3}\boldsymbol{e}_2\right) \\
&= \frac{1}{s}\left(\frac{\partial a_3}{\partial\phi}\boldsymbol{s}-\frac{\partial sa_\phi}{\partial x_3}\boldsymbol{s}-\frac{\partial a_s}{\partial\phi}\boldsymbol{e}_3-s\frac{\partial a_3}{\partial s}\boldsymbol{\phi}+\frac{\partial sa_\phi}{\partial s}\boldsymbol{e}_3+s\frac{\partial a_s}{\partial x_3}\boldsymbol{\phi}\right) \\
&= \frac{1}{s}\left(\frac{\partial a_3}{\partial\phi}-s\frac{\partial a_\phi}{\partial x_3}\right)\boldsymbol{s}+\left(\frac{\partial a_s}{\partial x_3}-\frac{\partial a_3}{\partial s}\right)\boldsymbol{\phi}+\frac{1}{s}\left(\frac{\partial sa_\phi}{\partial s}-\frac{\partial a_s}{\partial\phi}\right)\boldsymbol{e}_3
\end{aligned}
\tag{1.54}
$$

拉普拉斯算子

$$
\begin{aligned}
\boldsymbol{\nabla}^2 f &= \frac{1}{h_1h_2h_3}\frac{\partial}{\partial u_i}\left(h_1h_2h_3\frac{\partial f}{h_i^2\partial u_i}\right) \\
&= \frac{1}{s}\frac{\partial}{\partial u_i}\left(s\frac{\partial f}{h_i^2\partial u_i}\right) \\
&= \frac{1}{s}\frac{\partial}{\partial u_s}(s\frac{\partial f}{h_s^2\partial u_s})+\frac{1}{s}\frac{\partial}{\partial\phi}(s\frac{\partial f}{h_\phi^2\partial\phi})+\frac{1}{s}\frac{\partial}{\partial u_3}(s\frac{\partial f}{h_3^2\partial u_3}) \\
&= \frac{1}{s}\frac{\partial}{\partial u_s}(s\frac{\partial f}{\partial u_s})+\frac{1}{s}\frac{\partial}{\partial\phi}(s\frac{\partial f}{s^2\partial\phi})+\frac{1}{s}\frac{\partial}{\partial u_3}(s\frac{\partial f}{\partial u_3}) \\
&= \frac{1}{s}\frac{\partial}{\partial u_s}(s\frac{\partial f}{\partial u_s})+\frac{1}{s^2}\frac{\partial^2 f}{\partial\phi^2}+\frac{\partial^2 f}{\partial u_3^2}
\end{aligned}
\tag{1.55}
$$

1.4.2 球坐标系

将球坐标系坐标转化为直角系坐标的方式为

$$
x_1 = r\cos\phi\sin\theta,\ x_2 = r\sin\phi\sin\theta,\ x_3 = r\cos\theta \tag{1.56}
$$

球坐标系和直角系坐标无限小元转换方式为

$$
\begin{pmatrix}\mathrm{d}x_1\\ \mathrm{d}x_2\\ \mathrm{d}x_3\end{pmatrix}=\begin{pmatrix}\frac{x_1}{r} & -x_2 & \frac{x_1x_3}{r\sin\theta}\\ \frac{x_2}{r} & x_1 & \frac{x_2x_3}{r\sin\theta}\\ \frac{x_3}{r} & 0 & -r\sin\theta\end{pmatrix}\begin{pmatrix}\mathrm{d}r\\ \mathrm{d}\phi\\ \mathrm{d}\theta\end{pmatrix} \tag{1.57}
$$

其拉梅系数为

$$
h_1 = 1,\ h_2 = r\sin\theta,\ h_3 = r \tag{1.58}
$$

其无限小位移为

$$
\mathrm{d}\boldsymbol{l} = \mathrm{d}r\boldsymbol{r} + r\sin\theta\,\mathrm{d}\phi\boldsymbol{\phi} + r\,\mathrm{d}\theta\boldsymbol{\theta} \tag{1.59}
$$

其体积元为

$$
\mathrm{d}V = r\sin\theta r\,\mathrm{d}r\,\mathrm{d}\phi\,\mathrm{d}\theta \tag{1.60}
$$

梯度

$$
\nabla f = \frac{1}{h_i}\frac{\partial f}{\partial u_i}\boldsymbol{e}_i = \frac{\partial}{\partial r}\boldsymbol{r} + \frac{1}{r\sin\theta}\frac{\partial}{\partial\phi}\boldsymbol{\phi} + \frac{1}{r}\frac{\partial}{\partial\theta}\boldsymbol{\theta} \tag{1.61}
$$

散度

$$
\begin{aligned}
\boldsymbol{\nabla}\cdot\boldsymbol{a} &= \frac{1}{r^2\sin\theta}\left(\frac{\partial r^2\sin\theta a_r}{\partial r}+\frac{\partial r a_\phi}{\partial\phi}+\frac{\partial r\sin\theta a_\theta}{\partial\theta}\right)\\
&= \frac{1}{r^2\sin\theta}\left(\sin\theta\frac{\partial r^2 a_r}{\partial r}+r\frac{\partial a_\phi}{\partial\phi}+r\frac{\partial\sin\theta a_\theta}{\partial\theta}\right)\\
&= \frac{1}{r^2}\frac{\partial r^2 a_r}{\partial r}+\frac{1}{r\sin\theta}\frac{\partial a_\phi}{\partial\phi}+\frac{1}{r\sin\theta}\frac{\partial\sin\theta a_\theta}{\partial\theta}
\end{aligned}
\tag{1.62}
$$

旋度

$$
\begin{aligned}
&\boldsymbol{\nabla}\times\boldsymbol{a} \qquad (1.63)\\
=&\frac{h_i}{h_1h_2h_3}\varepsilon_{ijk}\frac{\partial(h_ka_k)}{\partial u_j}\boldsymbol{e}_i\\
=&\frac{h_i}{r^2\sin\theta}\varepsilon_{ijk}\frac{\partial(h_ka_k)}{\partial u_j}\boldsymbol{e}_i\\
=&\frac{1}{r^2\sin\theta}\left(\varepsilon_{123}\frac{\partial ra_3}{\partial u_2}\boldsymbol{e}_1+\varepsilon_{132}\frac{\partial sa_2}{\partial u_3}\boldsymbol{e}_1+r\varepsilon_{321}\frac{\partial a_1}{\partial u_2}\boldsymbol{e}_3+s\varepsilon_{213}\frac{\partial ra_3}{\partial u_1}\boldsymbol{e}_2+r\varepsilon_{312}\frac{\partial sa_2}{\partial u_1}\boldsymbol{e}_3+s\varepsilon_{231}\frac{\partial a_1}{\partial u_3}\boldsymbol{e}_2\right)\\
=&\frac{1}{r^2\sin\theta}\left(\frac{\partial ra_3}{\partial u_2}\boldsymbol{e}_1-\frac{\partial r\sin\theta a_2}{\partial u_3}\boldsymbol{e}_1-r\frac{\partial a_1}{\partial u_2}\boldsymbol{e}_3-r\sin\theta\frac{\partial ra_3}{\partial u_1}\boldsymbol{e}_2+r\frac{\partial r\sin\theta a_2}{\partial u_1}\boldsymbol{e}_3+r\sin\theta\frac{\partial a_1}{\partial u_3}\boldsymbol{e}_2\right)\\
=&\frac{1}{r^2\sin\theta}\left(\frac{\partial ra_\theta}{\partial\phi}\boldsymbol{r}-\frac{\partial r\sin\theta a_\phi}{\partial\theta}\boldsymbol{r}-r\frac{\partial a_r}{\partial\phi}\boldsymbol{\theta}-r\sin\theta\frac{\partial ra_\theta}{\partial r}\boldsymbol{\phi}+r\frac{\partial r\sin\theta a_\phi}{\partial r}\boldsymbol{\theta}+r\sin\theta\frac{\partial a_r}{\partial\theta}\boldsymbol{\phi}\right)\\
=&\frac{1}{r^2\sin\theta}\left(r\frac{\partial a_\theta}{\partial\phi}-r\frac{\partial\sin\theta a_\phi}{\partial\theta}\right)\boldsymbol{r}+\frac{1}{r}\left(\frac{\partial a_r}{\partial\theta}-\frac{\partial ra_\theta}{\partial r}\right)\boldsymbol{\phi}+\frac{1}{r\sin\theta}\left(\sin\theta\frac{\partial ra_\phi}{\partial r}-\frac{\partial a_r}{\partial\phi}\right)\boldsymbol{\theta}\\
=&\frac{1}{r\sin\theta}\left(\frac{\partial a_\theta}{\partial\phi}-\frac{\partial\sin\theta a_\phi}{\partial\theta}\right)\boldsymbol{r}+\frac{1}{r}\left(\frac{\partial a_r}{\partial\theta}-\frac{\partial ra_\theta}{\partial r}\right)\boldsymbol{\phi}+\frac{1}{r\sin\theta}\left(\sin\theta\frac{\partial ra_\phi}{\partial r}-\frac{\partial a_r}{\partial\phi}\right)\boldsymbol{\theta} \qquad (1.64)
\end{aligned}
$$

拉普拉斯算子

$$
\begin{aligned}
\boldsymbol{\nabla}^2 f &= \frac{1}{h_1h_2h_3}\frac{\partial}{\partial u_i}\left(h_1h_2h_3\frac{\partial f}{h_i^2\partial u_i}\right)\\
&= \frac{1}{r^2\sin\theta}\frac{\partial}{\partial u_i}\left(r^2\sin\theta\frac{\partial f}{h_i^2\partial u_i}\right)\\
&= \frac{1}{r^2\sin\theta}\frac{\partial}{\partial r}\left(r^2\sin\theta\frac{\partial f}{\partial r}\right)+\frac{1}{r^2\sin\theta}\frac{\partial}{\partial\phi}\left(\frac{\partial f}{\sin\theta\partial\phi}\right)+\frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial f}{\partial\theta}\right)\\
&= \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial f}{\partial r}\right)+\frac{1}{r^2\sin^2\theta}\frac{\partial^2 f}{\partial\phi^2}+\frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial f}{\partial\theta}\right)
\end{aligned}
\tag{1.65}
$$

1.5 狄拉克函数与阶跃函数

1.5.1 狄拉克函数

严格来说, 狄拉克 δ 函数并不是一个通常意义下的函数, 而是通过其在积分中的作用来定义的。设 $f(x)$ 是在 $x=0$ 附近连续, 并且在无穷远处足够快衰减的函数。定义狄拉克 δ 满足

$$
\int_{-\infty}^{\infty}\delta(x)\,f(x)\,\mathrm{d}x=f(0), \tag{1.66}
$$

由定义立即得到

$$
\int_{-\infty}^{\infty}\delta(x-a)\,f(x)\,\mathrm{d}x=f(a). \tag{1.67}
$$

设 $g(x)$ 在 x_i 处有孤立零点, 且 $g'(x_i)\neq 0$, 则

$$
\delta(g(x))=\sum_i\frac{\delta(x-x_i)}{|g'(x_i)|}. \tag{1.68}
$$

1.5.2 阶跃函数

定义阶跃函数

$$H(x)=\begin{cases}0, & x<0,\\ 1, & x>0.\end{cases} \tag{1.69}$$

1.6 阶跃函数的导数

在通常意义下, $H(x)$ 在 $x=0$ 不可导。但在积分意义下, 我们定义

$$\int_{-\infty}^{\infty}\frac{\mathrm{d}H(x)}{\mathrm{d}x}f(x)\,\mathrm{d}x\equiv-\int_{-\infty}^{\infty}H(x)\frac{\mathrm{d}f}{\mathrm{d}x}\,\mathrm{d}x. \tag{1.70}$$

对右侧分部积分：

$$\begin{aligned}-\int_{-\infty}^{\infty}H(x)\frac{\mathrm{d}f}{\mathrm{d}x}\,\mathrm{d}x&=-\int_{0}^{\infty}\frac{\mathrm{d}f}{\mathrm{d}x}\,\mathrm{d}x\\&=f(0).\end{aligned} \tag{1.71}$$

于是得到

$$\frac{\mathrm{d}H(x)}{\mathrm{d}x}=\delta(x), \tag{1.72}$$

1.6.1 多维狄拉克函数

三维 δ 函数定义为

$$\delta^{(3)}(\boldsymbol{r})=\delta(x)\delta(y)\delta(z), \tag{1.73}$$

并满足

$$\iiint\delta^{(3)}(\boldsymbol{r}-\boldsymbol{r}_0)f(\boldsymbol{r})\,\mathrm{d}V=f(\boldsymbol{r}_0). \tag{1.74}$$

在正交曲线坐标 (u_1,u_2,u_3) 中,

$$\mathrm{d}V=h_1h_2h_3\,\mathrm{d}u_1\,\mathrm{d}u_2\,\mathrm{d}u_3,$$

因此定义

$$\delta^{(3)}(\boldsymbol{r}-\boldsymbol{r}_0)=\frac{\delta(u_1-u_{1,0})\delta(u_2-u_{2,0})\delta(u_3-u_{3,0})}{h_1h_2h_3}. \tag{1.75}$$

1.7 矢量场理论

1.7.1 亥姆霍兹定理

设矢量场 $\boldsymbol{F}(\boldsymbol{r})$ 定义在整个空间中, 其散度与旋度分别给定为

$$\boldsymbol{\nabla}\cdot\boldsymbol{F}(\boldsymbol{r})=D(\boldsymbol{r}),\qquad\boldsymbol{\nabla}\times\boldsymbol{F}(\boldsymbol{r})=\boldsymbol{C}(\boldsymbol{r}).$$

为了使体积分有意义, 仅要求 $D(\boldsymbol{r})$ 与 $\boldsymbol{C}(\boldsymbol{r})$ 在无穷远处趋于零是不够的, 它们的衰减速度必须足够快。

考虑定义在整个空间上的积分

$$\iiint_{\mathbb{V}}X(\boldsymbol{r})\,\mathrm{d}V,$$

其中 $X(\boldsymbol{r})$ 表示 $D(\boldsymbol{r})$ 或 $\boldsymbol{C}(\boldsymbol{r})$ 的某一分量。

将积分写为球坐标形式,

$$\iiint_{\mathbb{V}} X(\boldsymbol{r})\,\mathrm{d}V = \int_0^\infty \left(\iint_S X(r,\Omega) r^2\,\mathrm{d}\Omega\right)\mathrm{d}r.$$

若存在常数 $A>0$ 及 $R>0$, 使得当 $r>R$ 时对所有方向 Ω 成立

$$|X(r,\Omega)| \geq \frac{A}{r},$$

则有下界估计

$$\iiint_{|\boldsymbol{r}|>R} |X(\boldsymbol{r})|\,\mathrm{d}V \geq A\int_R^\infty r\,\mathrm{d}r \iint_S \mathrm{d}\Omega,$$

该积分显然发散。

类似地, 若

$$|X(r,\Omega)| \geq \frac{A}{r^2},$$

则径向积分包含

$$\int_R^\infty \frac{\mathrm{d}r}{r},$$

从而发散。

因此, 为保证体积分收敛, 必须要求

$$X(\boldsymbol{r}) = o\left(\frac{1}{r^2}\right), \qquad r\to\infty, \tag{1.76}$$

该条件需在角向上一致成立。

因此, 为保证积分收敛, 必须要求

$$D(\boldsymbol{r}),\ \boldsymbol{C}(\boldsymbol{r}) = o\left(\frac{1}{r^2}\right), \qquad r\to\infty. \tag{1.77}$$

这一条件同时也足以保证在无穷远处的曲面积分为零。

设矢量场 $\boldsymbol{F}(\boldsymbol{r})$ 具有散度 $D(\boldsymbol{r})$ 与旋度 $\boldsymbol{C}(\boldsymbol{r})$。若取另一矢量场

$$\boldsymbol{F}'(\boldsymbol{r}) = \boldsymbol{F}(\boldsymbol{r}) + \boldsymbol{G}(\boldsymbol{r}),$$

且 $\boldsymbol{G}(\boldsymbol{r})$ 满足

$$\boldsymbol{\nabla}\cdot\boldsymbol{G} = 0, \qquad \boldsymbol{\nabla}\times\boldsymbol{G} = \boldsymbol{0},$$

则 $\boldsymbol{F}'$ 与 $\boldsymbol{F}$ 具有相同的散度与旋度。

因此, 仅给定散度与旋度并不能唯一确定矢量场。然而可以证明：不存在一个非零的矢量场 (见第三章) $\boldsymbol{G}(\boldsymbol{r})$, 它在整个空间中同时满足

$$\boldsymbol{\nabla}\cdot\boldsymbol{G} = 0, \qquad \boldsymbol{\nabla}\times\boldsymbol{G} = \boldsymbol{0},$$

并且在无穷远处趋于零。

因此, 若进一步要求

$$\boldsymbol{F}(\boldsymbol{r}) \to \boldsymbol{0} \qquad 当 r\to\infty,$$

则满足给定散度与旋度的矢量场解是唯一的。

现在可以严谨地表述亥姆霍兹定理如下：

> 若矢量场 $\boldsymbol{F}(\boldsymbol{r})$ 的散度 $D(\boldsymbol{r})$ 与旋度 $\boldsymbol{C}(\boldsymbol{r})$ 已知, 且二者在 $r\to\infty$ 时均比 $1/r^2$ 衰减得更快, 同时 $\boldsymbol{F}(\boldsymbol{r})$ 在无穷远处趋于零, 则 $\boldsymbol{F}(\boldsymbol{r})$ 被其散度与旋度唯一确定。

1.7.2 势函数

1. 标量势与无旋场 设向量场 $\boldsymbol{F}(\boldsymbol{r})$ 定义在某一区域 $V \subset \mathbb{R}^3$。若存在标量函数 $\phi(\boldsymbol{r})$, 使得

$$\boldsymbol{F} = -\boldsymbol{\nabla}\phi, \tag{1.78}$$

则称 ϕ 为 $\boldsymbol{F}$ 的**标量势函数**, $\boldsymbol{F}$ 称为**保守场**。由旋度的定义可得

$$\boldsymbol{\nabla} \times \boldsymbol{F} = -\boldsymbol{\nabla} \times \boldsymbol{\nabla}\phi = \boldsymbol{0}. \tag{1.79}$$

因此, 任何具有标量势的向量场必为无旋场。

2. 无旋场的势函数存在性 若

$$\boldsymbol{\nabla} \times \boldsymbol{F} = \boldsymbol{0}, \tag{1.80}$$

并且区域 V 是单连通的 (任意闭合曲线可连续收缩为一点), 则由斯托克斯定理可得

$$\oint_C \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{l} = \iint_S (\boldsymbol{\nabla} \times \boldsymbol{F}) \cdot \mathrm{d}\boldsymbol{S} = 0, \tag{1.81}$$

线积分与路径无关, 可定义

$$\phi(\boldsymbol{r}) = -\int_{\boldsymbol{r}_0}^{\boldsymbol{r}} \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{l}, \tag{1.82}$$

从而得到 $\boldsymbol{F} = -\boldsymbol{\nabla}\phi$。

3. 向量势与无散场 若存在向量函数 $\boldsymbol{A}(\boldsymbol{r})$, 使

$$\boldsymbol{F} = \boldsymbol{\nabla} \times \boldsymbol{A}, \tag{1.83}$$

则称 $\boldsymbol{A}$ 为 $\boldsymbol{F}$ 的**向量势**。由恒等式

$$\boldsymbol{\nabla} \cdot (\boldsymbol{\nabla} \times \boldsymbol{A}) = 0 \tag{1.84}$$

可知, 任何具有向量势的向量场必为无散场。

4. 势函数的不唯一性

- 若 $\boldsymbol{F} = -\boldsymbol{\nabla}\phi$, 则对任意常数 C, $\phi' = \phi + C$ 给出同一向量场。
- 若 $\boldsymbol{F} = \boldsymbol{\nabla} \times \boldsymbol{A}$, 则对任意标量函数 $\chi(\boldsymbol{r})$, $\boldsymbol{A}' = \boldsymbol{A} + \boldsymbol{\nabla}\chi$ 产生相同的 $\boldsymbol{F}$(**规范变换**)。

5. 亥姆霍兹分解 在满足适当衰减条件的情况下, 任意向量场 $\boldsymbol{F}$ 可分解为

$$\boldsymbol{F} = -\boldsymbol{\nabla}\phi + \boldsymbol{\nabla} \times \boldsymbol{A}, \tag{1.85}$$

其中

$$\boldsymbol{\nabla}^2\phi = -\boldsymbol{\nabla} \cdot \boldsymbol{F}, \tag{1.86}$$

$$\boldsymbol{\nabla}^2\boldsymbol{A} = -\boldsymbol{\nabla} \times \boldsymbol{F}. \tag{1.87}$$

这就是亥姆霍兹定理在势函数语言下的表达。

1.8 习题

1.8.1 证明: $\boldsymbol{a}\times(\boldsymbol{b}\times\boldsymbol{c})+\boldsymbol{c}\times(\boldsymbol{a}\times\boldsymbol{b})+\boldsymbol{b}\times(\boldsymbol{c}\times\boldsymbol{a})=0$

$$\boldsymbol{a}\times(\boldsymbol{b}\times\boldsymbol{c})+\boldsymbol{c}\times(\boldsymbol{a}\times\boldsymbol{b})+\boldsymbol{b}\times(\boldsymbol{c}\times\boldsymbol{a})=-(\boldsymbol{b}\cdot\boldsymbol{a})\boldsymbol{c}+(\boldsymbol{a}\cdot\boldsymbol{b})\boldsymbol{c}-(\boldsymbol{a}\cdot\boldsymbol{c})\boldsymbol{b}+(\boldsymbol{b}\cdot\boldsymbol{c})\boldsymbol{a}-(\boldsymbol{c}\cdot\boldsymbol{b})\boldsymbol{a}+(\boldsymbol{a}\cdot\boldsymbol{b})\boldsymbol{c}=0$$

1.8.2 证明: $\boldsymbol{Ra}\cdot\boldsymbol{Rb}=\boldsymbol{a}\cdot\boldsymbol{b}$, 其中 $\boldsymbol{R}$ 为旋转矩阵

$$\boldsymbol{Ra}\cdot\boldsymbol{Rb}=(\boldsymbol{Ra})^T(\boldsymbol{Rb})=\boldsymbol{a}^T\boldsymbol{R}^T\boldsymbol{Rb}=\boldsymbol{a}^T\boldsymbol{R}^{-1}\boldsymbol{Rb}=\boldsymbol{a}^T\boldsymbol{b}=\boldsymbol{a}\cdot\boldsymbol{b}$$

1.8.3 在坐标逆变换 $(x'=-x,\ y'=-y,\ z'=-z)$ 下, 两个矢量的叉乘是如何变换的?

$$\boldsymbol{a}'\times\boldsymbol{b}'=\begin{pmatrix}\boldsymbol{e}_i & \boldsymbol{e}_j & \boldsymbol{e}_k\\ a_i' & a_j' & a_k'\\ b_i' & b_j' & b_k'\end{pmatrix}=\begin{pmatrix}\boldsymbol{e}_i & \boldsymbol{e}_j & \boldsymbol{e}_k\\ -a_i & -a_j & -a_k\\ -b_i & -b_j & -b_k\end{pmatrix}=\boldsymbol{a}\times\boldsymbol{b}$$

1.8.4 在坐标逆变换下, 标量三重积 $(\boldsymbol{c}\cdot(\boldsymbol{a}\times\boldsymbol{b}))$ 是如何变换的?

$$\boldsymbol{c}'\cdot(\boldsymbol{a}'\times\boldsymbol{b}')=\boldsymbol{c}'\cdot(\boldsymbol{a}\times\boldsymbol{b})=-\boldsymbol{c}\cdot(\boldsymbol{a}\times\boldsymbol{b})$$

1.8.5 $\boldsymbol{n}$ 是固定点 (x',y',z') 到点 (x,y,z) 的间隔矢量, n 是它的长度, 证明:

(a) $\boldsymbol{\nabla}n^2=2\boldsymbol{n}$.

(b) $\boldsymbol{\nabla}\frac{1}{n}=-\frac{\boldsymbol{n}}{n^2}$.

(c) $\boldsymbol{\nabla}n^a=a\boldsymbol{n}n^{a-2}(a\neq 0)$.

(a) $\boldsymbol{\nabla}n^2=\frac{\partial n^2}{\partial e_i}\boldsymbol{e}_i=2n\frac{\partial n}{\partial e_i}\boldsymbol{e}_i=2\boldsymbol{n}$

(c) $\boldsymbol{\nabla}n^a=\frac{\partial n^a}{\partial e_i}\boldsymbol{e}_i=an^{-1}\frac{\partial n}{\partial e_i}\boldsymbol{e}_i=a\boldsymbol{n}n^{a-2}$

1.8.6 函数 f 只依赖平面坐标 y,z。令旋转后的坐标记为 $\begin{pmatrix}y'\\ z'\end{pmatrix}=\begin{pmatrix}\cos\theta & \sin\theta\\ -\sin\theta & \cos\theta\end{pmatrix}\begin{pmatrix}y\\ z\end{pmatrix}$, 求证 $\boldsymbol{\nabla}f'=\begin{pmatrix}\cos\theta & \sin\theta\\ -\sin\theta & \cos\theta\end{pmatrix}\begin{pmatrix}\frac{\partial f}{\partial y}\\ \frac{\partial f}{\partial z}\end{pmatrix}$

$$\begin{pmatrix}y\\ z\end{pmatrix}=\begin{pmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{pmatrix}\begin{pmatrix}y'\\ z'\end{pmatrix}$$

$$\frac{\partial f}{\partial y'}=\frac{\partial f}{\partial y}\frac{\partial y}{\partial y'}+\frac{\partial f}{\partial z}\frac{\partial z}{\partial y'}=\frac{\partial f}{\partial y}\cos\theta+\frac{\partial f}{\partial z}\sin\theta$$

$$\frac{\partial f}{\partial z'}=\frac{\partial f}{\partial y}\frac{\partial y}{\partial z'}+\frac{\partial f}{\partial z}\frac{\partial z}{\partial z'}=-\frac{\partial f}{\partial y}\sin\theta+\frac{\partial f}{\partial z}\cos\theta$$

$$\begin{pmatrix}\dfrac{\partial f}{\partial y'}\\ \dfrac{\partial f}{\partial z'}\end{pmatrix}=\begin{pmatrix}\cos\theta & \sin\theta\\ -\sin\theta & \cos\theta\end{pmatrix}\begin{pmatrix}\dfrac{\partial f}{\partial y}\\ \dfrac{\partial f}{\partial z}\end{pmatrix}$$

1.8.7 计算 $(\boldsymbol{\nabla T})\times(\boldsymbol{\nabla S})$

$$
\begin{aligned}
(\boldsymbol{\nabla T})\times(\boldsymbol{\nabla S}) &= \left(\varepsilon_{ijk}\frac{\partial T_j}{\partial x_i}\boldsymbol{e}_k\right)\times\left(\varepsilon_{lmn}\frac{\partial S_m}{\partial x_l}\boldsymbol{e}_n\right)\\
&= \varepsilon_{opq}\varepsilon_{ijo}\frac{\partial T_j}{\partial x_i}\varepsilon_{lmp}\frac{\partial S_m}{\partial x_l}\boldsymbol{e}_q\\
&= \varepsilon_{pqo}\varepsilon_{ijo}\frac{\partial T_j}{\partial x_i}\varepsilon_{lmp}\frac{\partial S_m}{\partial x_l}\boldsymbol{e}_q\\
&= (\delta_{ip}\delta_{jq}-\delta_{iq}\delta_{jp})\frac{\partial T_j}{\partial x_i}\varepsilon_{lmp}\frac{\partial S_m}{\partial x_l}\boldsymbol{e}_q\\
&= \delta_{ip}\delta_{jq}\frac{\partial T_j}{\partial x_i}\varepsilon_{lmp}\frac{\partial S_m}{\partial x_l}\boldsymbol{e}_q-\delta_{iq}\delta_{jp}\frac{\partial T_j}{\partial x_i}\varepsilon_{lmp}\frac{\partial S_m}{\partial x_l}\boldsymbol{e}_q\\
&= \frac{\partial T_j}{\partial x_i}\varepsilon_{lmi}\frac{\partial S_m}{\partial x_l}\boldsymbol{e}_j-\frac{\partial T_j}{\partial x_i}\varepsilon_{lmj}\frac{\partial S_m}{\partial x_l}\boldsymbol{e}_i\\
&= \frac{\partial T_i}{\partial x_j}\varepsilon_{lmj}\frac{\partial S_m}{\partial x_l}\boldsymbol{e}_i-\frac{\partial T_j}{\partial x_i}\varepsilon_{lmj}\frac{\partial S_m}{\partial x_l}\boldsymbol{e}_i\\
&= \left(\frac{\partial T_i}{\partial x_j}-\frac{\partial T_j}{\partial x_i}\right)\varepsilon_{lmj}\frac{\partial S_m}{\partial x_l}\boldsymbol{e}_i
\end{aligned}
$$

1.8.8 证明 $\iint_S f(\boldsymbol{\nabla}\times\boldsymbol{a})\cdot\mathrm{d}\boldsymbol{S}=\iint_S[\boldsymbol{a}\times(\boldsymbol{\nabla}f)]\cdot\mathrm{d}\boldsymbol{S}+\oint_{\partial S}f\boldsymbol{a}\cdot\mathrm{d}\boldsymbol{l}$

$$
\begin{aligned}
\oint_{\partial S}f\boldsymbol{a}\cdot\mathrm{d}\boldsymbol{l} &= \iint_S[\boldsymbol{\nabla}\times(\boldsymbol{f}\boldsymbol{a})]\cdot\mathrm{d}\boldsymbol{S}\\
&= \iint_S[f(\boldsymbol{\nabla}\times\boldsymbol{a})+(\boldsymbol{\nabla}f)\times\boldsymbol{a}]\cdot\mathrm{d}\boldsymbol{S}\\
\iint_S f(\boldsymbol{\nabla}\times\boldsymbol{a})\cdot\mathrm{d}\boldsymbol{S} &= \iint_S[\boldsymbol{a}\times(\boldsymbol{\nabla}f)]\cdot\mathrm{d}\boldsymbol{S}+\oint_{\partial S}f\boldsymbol{a}\cdot\mathrm{d}\boldsymbol{l}
\end{aligned}
$$

1.8.9 证明 $\iiint_V\boldsymbol{b}\cdot(\boldsymbol{\nabla}\times\boldsymbol{a})\,\mathrm{d}\tau=\iiint_V\boldsymbol{a}\cdot(\boldsymbol{\nabla}\times\boldsymbol{b})\,\mathrm{d}\tau+\iint(\boldsymbol{a}\times\boldsymbol{b})\cdot\mathrm{d}\boldsymbol{S}_{\partial V}$

$$
\begin{aligned}
\iint_{\partial V}(\boldsymbol{a}\times\boldsymbol{b})\cdot\mathrm{d}\boldsymbol{S} &= \iiint_V\boldsymbol{\nabla}\cdot(\boldsymbol{a}\times\boldsymbol{b})\,\mathrm{d}V\\
&= \iiint_V[\boldsymbol{b}\cdot(\boldsymbol{\nabla}\times\boldsymbol{a})-\boldsymbol{a}\cdot(\boldsymbol{\nabla}\times\boldsymbol{b})]\,\mathrm{d}V\\
\iiint_V\boldsymbol{b}\cdot(\boldsymbol{\nabla}\times\boldsymbol{a})\,\mathrm{d}\tau &= \iiint_V\boldsymbol{a}\cdot(\boldsymbol{\nabla}\times\boldsymbol{b})\,\mathrm{d}\tau+\iint_{\partial V}(\boldsymbol{a}\times\boldsymbol{b})\cdot\mathrm{d}\boldsymbol{S}
\end{aligned}
$$

1.8.10 设 $f=f(r)$, 求 $\boldsymbol{\nabla}^2[f(r)]$

$$
\begin{aligned}
\boldsymbol{\nabla}^2[f(r)] &= \frac{1}{sr}\frac{\partial}{\partial r}\left(sr\frac{\partial f}{\partial r}\right)+\frac{1}{s^2}\frac{\partial}{\partial \phi}\left(\frac{\partial f}{\partial \phi}\right)+\frac{1}{sr^2}\frac{\partial}{\partial \theta}\left(s\frac{\partial f}{\partial \theta}\right)\\
&= \frac{1}{sr}\frac{\partial}{\partial r}\left(sr\frac{\partial f}{\partial r}\right)\\
&= \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial f}{\partial r}\right)\\
&= \frac{2}{r}\frac{\partial f}{\partial r}+\frac{\partial^2 f}{\partial r^2}
\end{aligned}
\tag{1.88}
$$

1.8.11 证明:$\iiint_V \boldsymbol{\nabla}\times\boldsymbol{a}\,\mathrm{d}V=\oiint \mathrm{d}\boldsymbol{S}\times\boldsymbol{a}$

取任意常向量 $\boldsymbol{c}$。利用式1.39得 $\boldsymbol{\nabla}\times\boldsymbol{c}=\boldsymbol{0}$, 因此

$$\boldsymbol{\nabla}\cdot(\boldsymbol{a}\times\boldsymbol{c})=\boldsymbol{c}\cdot(\boldsymbol{\nabla}\times\boldsymbol{a}).$$

应用高斯定理,

$$\iiint_V \boldsymbol{\nabla}\cdot(\boldsymbol{a}\times\boldsymbol{c})\,\mathrm{d}V=\oiint_{\partial V}(\boldsymbol{a}\times\boldsymbol{c})\cdot\mathrm{d}\boldsymbol{S}=\iiint_V \boldsymbol{c}\cdot(\boldsymbol{\nabla}\times\boldsymbol{a})\,\mathrm{d}V.$$

利用式1.12得

$$\boldsymbol{c}\cdot\iiint_V(\boldsymbol{\nabla}\times\boldsymbol{a})\,\mathrm{d}V=\boldsymbol{c}\cdot\oiint_{\partial V}\mathrm{d}\boldsymbol{S}\times\boldsymbol{a}.$$

由于上述等式对任意常向量 $\boldsymbol{c}$ 成立, 必有

$$\iiint_V(\boldsymbol{\nabla}\times\boldsymbol{a})\,\mathrm{d}V=\oiint_{\partial V}\mathrm{d}\boldsymbol{S}\times\boldsymbol{a}.$$

1.8.12 证明:$\iint_S \mathrm{d}\boldsymbol{S}\times\boldsymbol{\nabla}f=\oint_{\partial S} f\,\mathrm{d}\boldsymbol{l}$

取任意常向量 $\boldsymbol{c}$。利用式1.28得

$$\iint_S(\boldsymbol{\nabla}f\times\boldsymbol{c})\cdot\mathrm{d}\boldsymbol{S}=\iint_S(\boldsymbol{\nabla}\times(f\boldsymbol{c}))\cdot\mathrm{d}\boldsymbol{S}.$$

由斯托克斯定理

$$\iint_S(\boldsymbol{\nabla}\times\boldsymbol{A})\cdot\mathrm{d}\boldsymbol{S}=\oint_{\partial S}\boldsymbol{A}\cdot\mathrm{d}\boldsymbol{l},$$

取 $\boldsymbol{A}=f\boldsymbol{c}$, 得

$$\iint_S(\boldsymbol{\nabla}\times(f\boldsymbol{c}))\cdot\mathrm{d}\boldsymbol{S}=\oint_{\partial S}(f\boldsymbol{c})\cdot\mathrm{d}\boldsymbol{l}=\oint_{\partial S}f\,\boldsymbol{c}\cdot\mathrm{d}\boldsymbol{l}.$$

又由混合积恒等式

$$\boldsymbol{c}\cdot(\mathrm{d}\boldsymbol{S}\times\boldsymbol{\nabla}f)=(\boldsymbol{\nabla}f\times\boldsymbol{c})\cdot\mathrm{d}\boldsymbol{S},$$

所以

$$\boldsymbol{c}\cdot\iint_S\mathrm{d}\boldsymbol{S}\times\boldsymbol{\nabla}f=\boldsymbol{c}\cdot\oint_{\partial S}f\ \mathrm{d}\boldsymbol{l}.$$

由于上述等式对任意常向量 $\boldsymbol{c}$ 都成立, 必有

$$\iint_S\mathrm{d}\boldsymbol{S}\times\boldsymbol{\nabla}f=\oint_{\partial S}f\ \mathrm{d}\boldsymbol{l}.$$

1.8.13 证明:$\oiint_{\partial S}(\boldsymbol{a}\times\boldsymbol{r})\cdot\mathrm{d}\boldsymbol{l}=2\iint_S\boldsymbol{a}\cdot\mathrm{d}\boldsymbol{S}$

$$
\begin{aligned}
\oiint_{\partial S}(\boldsymbol{a}\times\boldsymbol{r})\cdot\mathrm{d}\boldsymbol{l}&=\iint_S[\boldsymbol{\nabla}\times(\boldsymbol{a}\times\boldsymbol{r})]\cdot\mathrm{d}\boldsymbol{S}\\
&=\iint_S[\boldsymbol{\nabla}\times(\varepsilon_{ijk}a_ix_j\boldsymbol{e}_k)]\cdot\mathrm{d}\boldsymbol{S}\\
&=\iint_S(\varepsilon_{lmn}\varepsilon_{ijm}\frac{\partial a_ix_j}{\partial x_l}\boldsymbol{e}_n)\cdot\mathrm{d}\boldsymbol{S}\\
&=-\iint_S(\varepsilon_{lnm}\varepsilon_{ijm}\frac{\partial a_ix_j}{\partial x_l}\boldsymbol{e}_n)\cdot\mathrm{d}\boldsymbol{S}\\
&=-\iint_S(\delta_{il}\delta_{jn}-\delta_{in}\delta_{jl})(\frac{\partial a_ix_j}{\partial x_l}\boldsymbol{e}_n)\cdot\mathrm{d}\boldsymbol{S}\\
&=\iint_S(\delta_{in}\delta_{jl}\frac{\partial a_ix_j}{\partial x_l}\boldsymbol{e}_n-\delta_{il}\delta_{jn}\frac{\partial a_ix_j}{\partial x_l}\boldsymbol{e}_n)\cdot\mathrm{d}\boldsymbol{S}\\
&=\iint_S(\frac{\partial a_ix_j}{\partial x_j}\boldsymbol{e}_i-\frac{\partial a_ix_j}{\partial x_i}\boldsymbol{e}_j)\cdot\mathrm{d}\boldsymbol{S}\\
&=\iint_S(\frac{\partial a_jx_i}{\partial x_i}\boldsymbol{e}_j-\frac{\partial a_ix_j}{\partial x_i}\boldsymbol{e}_j)\cdot\mathrm{d}\boldsymbol{S}\\
&=\iint_S(a_j\frac{\partial x_i}{\partial x_i}+x_i\frac{\partial a_j}{\partial x_i}-a_i\frac{\partial x_j}{\partial x_i}-x_j\frac{\partial a_i}{\partial x_i})\boldsymbol{e}_j\cdot\mathrm{d}\boldsymbol{S}\\
&=\iint_S(3a_j+x_i\frac{\partial a_j}{\partial x_i}-a_i\delta_{ij}-x_j\frac{\partial a_i}{\partial x_i})\boldsymbol{e}_j\cdot\mathrm{d}\boldsymbol{S}\\
&=\iint_S 2a_j\boldsymbol{e}_j\cdot\mathrm{d}\boldsymbol{S}\\
&=\iint_S\boldsymbol{a}\cdot\mathrm{d}\boldsymbol{S}
\end{aligned}
$$

1.8.14 将下列电荷分布表示成电荷密度函数 ρ:1. 电荷量 Q 均匀分布在半径为 R 的球面上; 2. 电荷均匀分布在半径为 R 的柱面上, 单位长度的电荷量为 λ; 3. 电荷量均匀分布在半径为 R 的平面圆盘上

$$
\begin{aligned}
&1.\rho=\delta(R\boldsymbol{e}_r-\boldsymbol{r})\frac{Q}{4\pi R^2}\\
&2.\rho=\delta(R\boldsymbol{e}_s-\boldsymbol{s})\frac{\lambda}{2\pi R}\\
&3.\rho=\delta(\boldsymbol{x}_3)H(x+R)H(-x-R)\frac{\lambda}{\pi R^2}
\end{aligned}
$$

1.8.15 对函数 $\boldsymbol{a} = r^2\cos\theta\boldsymbol{r} + r^2\cos\phi\boldsymbol{\theta} - r^2\cos\theta\sin\phi\boldsymbol{\phi}$ 验证散度定理, 体积为半径为 R 在第一卦限的 $1/8$ 球体

$$\boldsymbol{\nabla}\cdot\boldsymbol{a} = 2r\cos\theta - r^2\cos\theta\cos\phi$$

$$\begin{aligned}
\int_0^{\frac{\pi}{2}}\mathrm{d}\phi\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^R\boldsymbol{\nabla}\cdot\boldsymbol{a}\,\mathrm{d}r &= \int_0^{\frac{\pi}{2}}\mathrm{d}\phi\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^R\left(2r\cos\theta - r^2\cos\theta\cos\phi\right)\mathrm{d}r \\
&= \int_0^{\frac{\pi}{2}}\mathrm{d}\phi\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\left(r^2\cos\theta - \frac{r^3}{3}\cos\theta\cos\phi\right)\Big|_0^R \\
&= \int_0^{\frac{\pi}{2}}\mathrm{d}\phi\int_0^{\frac{\pi}{2}}\left(R^2\cos\theta - \frac{R^3}{3}\cos\theta\cos\phi\right)\mathrm{d}\theta \\
&= \int_0^{\frac{\pi}{2}}\mathrm{d}\phi\left(R^2\sin\theta - \frac{R^3}{3}\sin\theta\cos\phi\right)\Big|_0^{\frac{\pi}{2}} \\
&= \int_0^{\frac{\pi}{2}}\left(R^2 - \frac{R^3}{3}\cos\phi\right)\mathrm{d}\phi \\
&= \left(R^2\phi - \frac{R^3}{3}\sin\phi\right)\Big|_0^{\frac{\pi}{2}} \\
&= \frac{\pi}{2}R^2 - \frac{R^3}{3}
\end{aligned}$$

计算 XY 平面:

$$\begin{aligned}
\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^R a_\phi\,\mathrm{d}r &= -\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^R r^2\cos\theta\sin\phi\,\mathrm{d}r \\
&= -\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^R r^2\cos\theta\,\mathrm{d}r \\
&= -\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\,\frac{r^3}{3}\cos\theta\Big|_0^R \\
&= -\int_0^{\frac{\pi}{2}}\frac{R^3}{3}\cos\theta\,\mathrm{d}\theta \\
&= -\frac{R^3}{3}
\end{aligned}$$

计算 XZ 平面:

$$\begin{aligned}
\int_0^{\frac{\pi}{2}}\mathrm{d}\phi\int_0^R a_\theta\,\mathrm{d}r &= \int_0^{\frac{\pi}{2}}\mathrm{d}\phi\int_0^R r^2\cos\phi\,\mathrm{d}r \\
&= \int_0^{\frac{\pi}{2}}\mathrm{d}\phi\,\frac{r^3}{3}\cos\phi\Big|_0^R \\
&= \int_0^{\frac{\pi}{2}}\frac{R^3}{3}\cos\phi\,\mathrm{d}\phi \\
&= \frac{R^3}{3}
\end{aligned}$$

计算 YZ 平面:

$$\int_0^{\frac{\pi}{2}}\mathrm{d}\phi\int_0^R a_\theta\,\mathrm{d}r = \int_0^{\frac{\pi}{2}}\mathrm{d}\phi\int_0^R r^2\cos\phi\,\mathrm{d}r = \frac{R^3}{3}$$

1.8.16 对函数 $\boldsymbol{a}=bx_2\boldsymbol{e}_1+cx_1\boldsymbol{e}_2$ 验证斯托克斯定理, 面的边界线选为处在 XY 平面, 半径为 R, 圆心在原点的圆周线。

$$\boldsymbol{\nabla}\times\boldsymbol{a}=\boldsymbol{\nabla}\times(by\boldsymbol{e}_1+cx\boldsymbol{e}_2)=\begin{pmatrix}\boldsymbol{e}_1 & \boldsymbol{e}_2 & \boldsymbol{e}_3\\ \frac{\partial}{\partial x_1} & \frac{\partial}{\partial x_2} & \frac{\partial}{\partial x_3}\\ bx_2 & cx_1 & 0\end{pmatrix}=(c-b)\boldsymbol{e}_3$$

$$\iint\boldsymbol{\nabla}\times\boldsymbol{a}\cdot\mathrm{d}\boldsymbol{l}=\iint(c-b)\boldsymbol{e}_3\cdot\mathrm{d}\boldsymbol{S}=2\pi R(c-b)$$

$$\begin{aligned}\oint(bR\sin\phi\boldsymbol{e}_1+cR\cos\phi\boldsymbol{e}_2)\cdot\mathrm{d}\boldsymbol{l}=&\int_0^{2\pi}(bR\sin\phi\boldsymbol{e}_1+cR\cos\phi\boldsymbol{e}_2)\cdot(-R\sin\phi\boldsymbol{e}_1\ +R\cos\phi\boldsymbol{e}_2\ )\,\mathrm{d}\phi\\ =&\int_0^{2\pi}(-bR^2\sin^2\phi+cR^2\cos^2\phi)\,\mathrm{d}\phi\\ =&2\pi R(c-b)\end{aligned}$$

1.8.17 对函数 $\boldsymbol{a}=r\cos^2\theta\boldsymbol{r}-r\sin\theta\cos\theta\boldsymbol{\theta}+3r\boldsymbol{\phi}$ 验证斯托克斯定理。

$$\begin{aligned}&\boldsymbol{\nabla}\times\boldsymbol{a}\\ =&\frac{1}{sr}\left(r\frac{\partial a_\theta}{\partial\phi}-s\frac{\partial a_\phi}{\partial\theta}\right)\boldsymbol{r}+\frac{1}{r}\left(\frac{\partial a_r}{\partial\theta}-\frac{\partial ra_\theta}{\partial r}\right)\boldsymbol{\phi}+\frac{1}{s}\left(\frac{\partial sa_\phi}{\partial r}-\frac{\partial a_r}{\partial\phi}\right)\boldsymbol{\theta}\\ =&\frac{1}{sr}\left(r\frac{\partial r\sin\theta\cos\theta}{\partial\phi}-s\frac{\partial 3r}{\partial\theta}\right)\boldsymbol{r}+\frac{1}{r}\left(\frac{\partial r\cos^2\theta}{\partial\theta}-\frac{\partial rr\sin\theta\cos\theta}{\partial r}\right)\boldsymbol{\phi}+\frac{1}{s}\left(\frac{\partial s3r}{\partial r}-\frac{\partial r\cos^2\theta}{\partial\phi}\right)\boldsymbol{\theta}\\ =&\frac{1}{s}\left(6r\sin\theta\right)\boldsymbol{\theta}\\ =&6\boldsymbol{\theta}\end{aligned}$$

计算 XY 平面:

$$\begin{aligned}\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^R a_\phi\,\mathrm{d}r=&\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^R 6\,\mathrm{d}r\\ =&\frac{3}{2}R\end{aligned}$$

1.8.18 对函数 $\boldsymbol{a} = r^2\sin^2\theta\boldsymbol{r} + 4r^2\cos\theta\boldsymbol{\theta} + r^2\tan\theta\boldsymbol{\phi}$ 验证散度定理。

$$
\begin{aligned}
\boldsymbol{\nabla}\cdot\boldsymbol{a} &= \frac{1}{sr}(\frac{\partial sra_r}{\partial r} + r\frac{\partial a_\phi}{\partial\phi} + s\frac{\partial\theta}{\partial\theta}) \\
&= \frac{1}{sr}(\frac{\partial srr^2\sin^2\theta}{\partial r} + r\frac{\partial 4r^2\cos\theta}{\partial\phi} + s\frac{\partial r^2\tan\theta}{\partial\theta}) \\
&= \frac{1}{sr}(4sr^2\sin^2\theta + sr^2\frac{1}{\cos^2\theta}) \\
&= 4r\sin^2\theta + r\frac{1}{\cos^2\theta}
\end{aligned}
$$

$$
\begin{aligned}
&\int_0^R r^2\,\mathrm{d}r\int_0^{2\pi}\mathrm{d}\phi\int_0^{\frac{\pi}{6}}(4r\sin^2\theta + r\frac{1}{\cos^2\theta})\sin\theta\,\mathrm{d}\theta \\
=&\int_0^R 2\pi r^3\,\mathrm{d}r\int_0^{\frac{\pi}{6}}(4\sin^2\theta + \frac{1}{\cos^2\theta})\,\mathrm{d}\theta \\
=&\int_0^R 2\pi r^3\ (2\theta - \sin 2\theta + \tan\theta)|_0^{\frac{\pi}{6}}\ \mathrm{d}r \\
=&\int_0^R 2\pi r^3(\frac{\pi}{3} - \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2})\,\mathrm{d}r \\
=&\int_0^R 2\pi r^3\frac{\pi}{3}\,\mathrm{d}r \\
=&2\pi R^4\frac{\pi}{12}
\end{aligned}
$$

1.8.19 证明 $\iiint_V\left[f\boldsymbol{\nabla}^2 g + (\boldsymbol{\nabla}f)\cdot(\boldsymbol{\nabla}g)\right]\mathrm{d}V = \oiint_S(f\boldsymbol{\nabla}g)\cdot\mathrm{d}S$

$$
\begin{aligned}
\iiint_V(\boldsymbol{\nabla}\cdot\boldsymbol{a})\,\mathrm{d}V &= \iint_{\partial V}\boldsymbol{a}\cdot\mathrm{d}\boldsymbol{S} \\
\iiint_V[\boldsymbol{\nabla}\cdot(f\boldsymbol{\nabla}g)]\,\mathrm{d}V &= \iint_{\partial V}(f\boldsymbol{\nabla}g)\cdot\mathrm{d}\boldsymbol{S} \\
\iiint_V\left[f\boldsymbol{\nabla}^2 g + (\boldsymbol{\nabla}f)\cdot(\boldsymbol{\nabla}g)\right]\mathrm{d}V &= \iint_{\partial V}(f\boldsymbol{\nabla}g)\cdot\mathrm{d}\boldsymbol{S}
\end{aligned}
$$

1.8.20 求 $\boldsymbol{a}=r^n\boldsymbol{e}_r$ 的散度

$$
\begin{aligned}
\boldsymbol{\nabla}\cdot\boldsymbol{a} &= \frac{\partial a_i}{\partial x_i}\\
&= \frac{\partial x_i r^{n-1}}{\partial x_i}\\
&= 3r^{n-1}+x_i(n-1)r^{n-2}\frac{\partial r}{\partial x_i}\\
&= 3r^{n-1}+x_i(n-1)r^{n-2}\frac{x_i}{r}\\
&= 3r^{n-1}+(n-1)r^{n-1}\\
&= (n+2)r^{n-1}
\end{aligned}
$$

$$
\iint_{\partial V}\boldsymbol{a}\cdot\mathrm{d}\boldsymbol{S}=\iint_{\partial V}r^n\boldsymbol{e}_r\cdot\mathrm{d}\boldsymbol{S}=4\pi r^2r^n=
$$

1.8.21 求 $\boldsymbol{a}=r^n\boldsymbol{e}_r$ 的旋度

$$
\begin{aligned}
\boldsymbol{\nabla}\times\boldsymbol{a} &= \varepsilon_{ijk}\frac{\partial a_j}{\partial x_i}\boldsymbol{e}_k\\
&= \varepsilon_{ijk}\frac{\partial r^{n-1}x_j}{\partial x_i}\boldsymbol{e}_k\\
&= \varepsilon_{ijk}x_j\frac{\partial r^{n-1}}{\partial x_i}\boldsymbol{e}_k\\
&= (n-1)\varepsilon_{ijk}x_jr^{n-2}\frac{\partial r}{\partial x_i}\boldsymbol{e}_k\\
&= (n-1)\varepsilon_{ijk}x_jr^{n-2}\frac{x_i}{r}\boldsymbol{e}_k\\
&= (n-1)r^{n-3}\varepsilon_{ijk}x_ix_j\boldsymbol{e}_k\\
&= 0
\end{aligned}
$$

1.8.22 以匀角速绕轴转动的抛物线形金属丝，其方程为 $x^2=4ay$。一质量为 m 的小环套在此金属丝上，可沿着金属丝无摩擦滑动。求小环在 x 方向的运动微分方程。

$$
\begin{aligned}
\frac{\partial\dfrac{mx^2\omega^2}{2}+mgy+\dfrac{m\dot{y}^2}{2}+\dfrac{m\dot{x}^2}{2}}{\partial t}&=0\\
\frac{\partial\dfrac{mx^2\omega^2}{2}+mg\dfrac{x^2}{4a}+\dfrac{mx^2\dot{x}^2}{8a^2}+\dfrac{m\dot{x}^2}{2}}{\partial t}&=0\\
x\dot{x}\omega^2+gx\frac{\dot{x}}{2a}+\frac{x\dot{x}^3}{4a^2}+\frac{x^2\dot{x}\ddot{x}}{4a^2}+mx\dot{x}&=0\\
x\omega^2+\frac{gx}{2a}+\frac{x\dot{x}^2}{4a^2}+\frac{x^2\ddot{x}}{4a^2}+mx&=0
\end{aligned}
$$

2 静电学

2.1 电场

2.1.1 库仑定律

设一个静止点电荷 q 距检验电荷 Q 的距离为 r, 那么它作用在 Q 上的力是

$$\boldsymbol{F} = \frac{1}{4\pi\varepsilon_0}\frac{qQ}{r^3}\boldsymbol{r} \tag{2.1}$$

常数 ε_0 称为真空介电常数, $\varepsilon_0 = 8.85 \times 10^{-12}\ \dfrac{\mathrm{C}^2}{\mathrm{N}\cdot\mathrm{m}^2}$

2.1.2 电场

$$\boldsymbol{E} \equiv \frac{\boldsymbol{F}}{Q} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^3}\boldsymbol{r} \tag{2.2}$$

2.1.3 电场强度通量

$\boldsymbol{\Phi}_E$ 为电场强度通量

$$\boldsymbol{\Phi}_E \equiv \iint_S \boldsymbol{E}\cdot\mathrm{d}\boldsymbol{S} \tag{2.3}$$

2.1.4 高斯定理

ρ 为电荷密度

$$\boldsymbol{\nabla}\cdot\boldsymbol{E} = \frac{1}{\varepsilon_0}\rho \tag{2.4}$$

2.1.5 两个带点为 q 的电荷, 相距 d 放置, 求垂直于连线中点且距离为 x_2 处的电场, 如果把一个换成 $-q$ 会怎样? 当 $y >> d$ 时会怎样?

$$\begin{aligned}\frac{1}{4\pi\varepsilon_0}\frac{q}{r^3}\boldsymbol{r}_1 + \frac{1}{4\pi\varepsilon_0}\frac{q}{r^3}\boldsymbol{r}_2 &= 2\frac{1}{4\pi\varepsilon_0}\frac{q}{r^3}x_2\boldsymbol{e}_2 \\ &= \frac{1}{2\pi\varepsilon_0}\frac{qx_2}{r^3}\boldsymbol{e}_2 \\ &= \frac{1}{2\pi\varepsilon_0}\frac{qx_2}{(x_2^2+\frac{d^2}{4})^{\frac{3}{2}}}\boldsymbol{e}_2\end{aligned}$$

$$\begin{aligned}\frac{1}{4\pi\varepsilon_0}\frac{q}{r^3}\boldsymbol{r}_1 + \frac{1}{4\pi\varepsilon_0}\frac{q}{r^3}\boldsymbol{r}_2 &= 2\frac{1}{4\pi\varepsilon_0}\frac{q}{r^3}\frac{d}{2}\boldsymbol{e}_1 \\ &= \frac{1}{4\pi\varepsilon_0}\frac{qd}{r^3}\boldsymbol{e}_1 \\ &\approx \frac{1}{4\pi\varepsilon_0}\frac{qd}{x_2^3}\boldsymbol{e}_1\end{aligned}$$

2.1.6 一个长度为 $2L$ 的细杆均匀带电, 电荷线密度为 λ, 求垂直于杆且与杆中心距离为 x_2 处的电场

$$
\begin{aligned}
\int_{-L}^{L} \frac{1}{4\pi\varepsilon_0}\frac{\lambda}{r^3}\boldsymbol{r}\,\mathrm{d}x &= \int_{-L}^{L} \frac{1}{4\pi\varepsilon_0}\frac{\lambda}{r^3}x_2\boldsymbol{e}_2\,\mathrm{d}x \\
&= \frac{\lambda}{4\pi\varepsilon_0}\boldsymbol{e}_2\int_{-L}^{L}\frac{x_2}{r^3}\,\mathrm{d}x \\
&= \frac{\lambda}{4\pi\varepsilon_0}\boldsymbol{e}_2\int_{-L}^{L}\frac{x_2}{(x^2+x_2^2)^{\frac{3}{2}}}\,\mathrm{d}x \\
&\xlongequal{x=x_2\tan\theta} \frac{\lambda}{4\pi\varepsilon_0}\boldsymbol{e}_2\int_{-\arctan\frac{L}{x_2}}^{\arctan\frac{L}{x_2}}\frac{x_2}{(x_2^2\tan\theta^2+x_2^2)^{\frac{3}{2}}}\,\mathrm{d}x_2\tan\theta \\
&= \frac{\lambda}{4\pi\varepsilon_0}\boldsymbol{e}_2\int_{-\arctan\frac{L}{x_2}}^{\arctan\frac{L}{x_2}}\frac{x_2^2\cos^3\theta}{x_2^3\cos^2\theta}\,\mathrm{d}\theta \\
&= \frac{\lambda}{4\pi\varepsilon_0}\boldsymbol{e}_2\int_{-\arctan\frac{L}{x_2}}^{\arctan\frac{L}{x_2}}\frac{\cos\theta}{x_2^2}\,\mathrm{d}\theta \\
&= \frac{\lambda}{4\pi\varepsilon_0}\boldsymbol{e}_2\left.\frac{\sin\theta}{x_2}\right|_{-\arctan\frac{L}{x_2}}^{\arctan\frac{L}{x_2}} \\
&= \frac{\lambda}{2\pi\varepsilon_0 x_2}\boldsymbol{e}_2\frac{L}{\sqrt{L^2+x_2^2}}
\end{aligned}
$$

2.1.7 一个长度为 L 的细杆均匀带电, 电荷线密度为 λ, 求杆一端上方距杆 y 处的电场

$$
\begin{aligned}
\int_0^L \frac{\lambda\boldsymbol{e}_y}{4\pi\varepsilon_0 r^2}\,\mathrm{d}x_2 &= \int_0^L \frac{\lambda\boldsymbol{e}_y}{4\pi\varepsilon_0(x_2+y)^2}\,\mathrm{d}x_2 \\
&= -\left.\frac{\lambda\boldsymbol{e}_y}{4\pi\varepsilon_0(x_2+y)}\right|_0^L \\
&= \frac{\lambda\boldsymbol{e}_y}{4\pi\varepsilon_0(L+y)} - \frac{\lambda\boldsymbol{e}_y}{4\pi\varepsilon_0 y} \\
&= \frac{\lambda\boldsymbol{e}_y L}{4\pi\varepsilon_0(L+y)y}
\end{aligned}
$$

2.1.8 一个边长为 $2L$ 的正方形线框均匀带电, 电荷线密度为 λ, 求线框中心上方距 x_3 处的电场

$$
4\frac{\lambda}{2\pi\varepsilon_0\sqrt{L^2+x_3^2}}\boldsymbol{e}_3\frac{L}{2\sqrt{2L^2+x_3^2}}\frac{x_3}{x_3\sqrt{L^2+x_3^2}^2} = \frac{\lambda}{\pi\varepsilon_0\sqrt{2L^2+x_3^2}}\boldsymbol{e}_3\frac{2Lx_3}{L^2+x_3^2}
$$

2.1.9 一个半径为 r 的圆线框均匀带电, 电荷线密度为 λ, 求线框中心上方距 x_3 处的电场

$$2\pi r\frac{\lambda}{4\pi\varepsilon_0}\boldsymbol{e}_3\frac{x_3}{\sqrt{x_3^2+r^2}}=\frac{\lambda}{2\varepsilon_0}\boldsymbol{e}_3\frac{x_3 r}{\sqrt{x_3^2+r^2}^3}$$

2.1.10 一个半径为 r 的圆片均匀带电, 电荷面密度为 σ, 求圆片中心上方距 x_3 处的电场

$$\begin{aligned}\int_0^r\frac{\sigma}{2\varepsilon_0}\boldsymbol{e}_3\frac{x_3x_1}{\sqrt{x_3^2+x_1^2}^3}\,\mathrm{d}x_1&=-\frac{\sigma}{2\varepsilon_0}\boldsymbol{e}_3\frac{x_3}{\sqrt{x_3^2+x_1^2}}\bigg|_0^r\\&=\frac{\sigma}{2\varepsilon_0}\boldsymbol{e}_3\frac{x_3}{\sqrt{x_3^2}}-\frac{\sigma}{2\varepsilon_0}\boldsymbol{e}_3\frac{x_3}{\sqrt{x_3^2+r^2}}\\&=\frac{\sigma}{2\varepsilon_0}\boldsymbol{e}_3-\frac{\sigma}{2\varepsilon_0}\boldsymbol{e}_3\frac{x_3}{\sqrt{x_3^2+r^2}}\end{aligned}$$

2.1.11 一个半径为 r 的球面均匀带电, 电荷面密度为 σ, 求圆球上方距 d 处的电场 (分 $d>r$ 和 $d<r$ 讨论)

$$\begin{aligned}\int_\pi^0\frac{\sigma}{2\varepsilon_0}\boldsymbol{e}_3\frac{(x_3+d)\sqrt{x_1^2+x_2^2}}{\sqrt{(x_3+d)^2+x_1^2+x_2^2}^3}\,\mathrm{d}r\phi&=\int_\pi^0\frac{\sigma}{2\varepsilon_0}\boldsymbol{e}_3\frac{(r\cos\phi+d)\sqrt{x_1^2+x_2^2}}{\sqrt{(r\cos\phi+d)^2+x_1^2+x_2^2}^3}\,\mathrm{d}r\phi\\&=\int_\pi^0\frac{\sigma}{2\varepsilon_0}\boldsymbol{e}_3\frac{(r\cos\phi+d)r\sin\phi}{\sqrt{r^2+2rd\cos\phi+d^2}^3}\,\mathrm{d}r\phi\\&=\int_0^\pi\frac{\sigma}{2\varepsilon_0}\boldsymbol{e}_3\frac{(r\cos\phi+d)r^2\sin\phi}{\sqrt{r^2+2rd\cos\phi+d^2}^3}\,\mathrm{d}\phi\end{aligned}$$

2.1.12 在某个区域电场可以写为 $\boldsymbol{E}=kr^3\boldsymbol{e_r}$, 求电荷密度和包含在半径为 R, 球心在原点的闭合球面内的总电荷

$$\rho=\varepsilon_0\boldsymbol{\nabla}\cdot\boldsymbol{E}=\varepsilon_0\frac{1}{sr}\frac{\partial srE_r}{\partial r}=\varepsilon_0\frac{1}{sr}\frac{\partial srr^3}{\partial r}=\varepsilon_0\frac{1}{sr}5sr^4=5\varepsilon_0r^3k$$

$$4\pi\int_0^R5\varepsilon_0r^3\,\mathrm{d}r=5\pi\varepsilon_0R^4k$$

2.2 电势

2.2.1 定义

$$V(\boldsymbol{r})\equiv-\int_O^r\boldsymbol{E}\cdot\mathrm{d}\boldsymbol{l}\tag{2.5}$$

其中 O 为预先设置的标准参考点, 通常为无限远处

2.2.2 一个半径为 R 的均匀带电球体, 总电荷为 q, 求电势 $V(\boldsymbol{r})$

$r \geq R$:

$$\int_r^\infty \frac{q}{4\pi\varepsilon_0 x^2}\,\mathrm{d}x = -\frac{q}{4\pi\varepsilon_0 x}\bigg|_r^\infty = \frac{q}{4\pi\varepsilon_0 r}$$

$r < R$:

$$\begin{aligned}\frac{q}{4\pi\varepsilon_0 R} + \int_r^R \frac{q}{4\pi\varepsilon_0 x^2}\frac{x^3}{R^3}\,\mathrm{d}x &= \frac{q}{4\pi\varepsilon_0 R} + \frac{qx^2}{4\pi\varepsilon_0 R^3}\bigg|_r^R \\ &= \frac{q}{4\pi\varepsilon_0 R} + \frac{qR^2}{4\pi\varepsilon_0 R^3} - \frac{qr^2}{4\pi\varepsilon_0 R^3} \\ &= \frac{q}{2\pi\varepsilon_0 R} - \frac{qr^2}{4\pi\varepsilon_0 R^3}\end{aligned}$$

2.2.3 一条均匀带电的无限长直线, 电荷线密度为 λ, 求电势 $V(\boldsymbol{r})$

$$V = \int_r^\infty \frac{\lambda}{2\pi r}\,\mathrm{d}x = \frac{\lambda}{2\pi}\ln(x)\bigg|_r^\infty = \frac{1}{2\pi}\ln(r)$$

2.2.4 一个长度为 $2L$ 的细杆均匀带电, 电荷线密度为 λ, 求垂直于杆且与杆中心距离为 h 处的电势

$$\begin{aligned}V &= \int_h^\infty \frac{\lambda}{4\pi\varepsilon_0}\frac{2L}{x_2\sqrt{L^2+x_2^2}}\,\mathrm{d}x_2 \\ &= -\frac{\lambda}{2\pi\varepsilon_0}\ln\left(\frac{\sqrt{x^2+L^2}+L}{x}\right)\bigg|_h^\infty \\ &= \frac{\lambda}{2\pi\varepsilon_0}\ln\left(\frac{\sqrt{h^2+L^2}+L}{h}\right)\end{aligned}$$

2.2.5 一个半径为 r 的圆片均匀带电, 电荷面密度为 σ, 求圆片中心上方距 h 处的电势

$$\begin{aligned}V &= \int_h^\infty \left(\frac{\sigma}{2\varepsilon_0} - \frac{\sigma}{2\varepsilon_0}\frac{x_3}{\sqrt{x_3^2+r^2}}\right)\mathrm{d}x_3 \\ &= \frac{\sigma}{2\varepsilon_0}x_3 - \frac{\sigma}{2\varepsilon_0}\sqrt{x_3^2+r^2}\bigg|_h^\infty \\ &= -\frac{\sigma}{2\varepsilon_0}\frac{r^2}{x_3+\sqrt{x_3^2+r^2}}\bigg|_h^\infty \\ &= \frac{\sigma}{2\varepsilon_0}\frac{r^2}{h+\sqrt{h^2+r^2}}\end{aligned}$$

2.2.6 一个半径为 r 的圆线框均匀带电, 电荷线密度为 λ, 求线框中心上方距 h 处的电势

$$V=\int_h^\infty \frac{\lambda}{2\varepsilon_0}\frac{x_3 r}{\sqrt{x_3^2+r^2}^3}\,\mathrm{d}x_3=-\frac{\lambda r}{2\varepsilon_0}\frac{1}{\sqrt{x_3^2+r^2}}\bigg|_h^\infty=\frac{\lambda r}{2\varepsilon_0}\frac{1}{\sqrt{h^2+r^2}}$$

2.2.7 一个尖角向下的圆锥体均匀带电, 电荷面密度为 σ, 圆锥高度等于半径为 h, 求 $z=h$ 处的电势

$$\begin{aligned}
V&=\int_{-h}^0 \frac{\lambda(h+x_3)}{2\varepsilon_0}\frac{1}{\sqrt{(h-x_3)^2+(h+x_3)^2}}\,\mathrm{d}x_3\\
&=\int_{-h}^0 \frac{\lambda(h+x_3)}{2\varepsilon_0}\frac{1}{\sqrt{2h^2+2x_3^2}}\,\mathrm{d}x_3\\
&=\frac{\lambda}{2\sqrt{2}\varepsilon_0}\int_{-h}^0 \frac{h}{\sqrt{h^2+x_3^2}}+\frac{x_3}{\sqrt{h^2+x_3^2}}\,\mathrm{d}x_3\\
&=h\ln\left(x_3+\sqrt{h^2+x_3^2}\right)+\sqrt{h^2+x_3^2}\bigg|_{-h}^0\frac{\lambda}{2\sqrt{2}\varepsilon_0}\\
&=\left[h\ln\left(\sqrt{h^2}\right)+\sqrt{h^2}-h\ln\left(-h+\sqrt{2h^2}\right)-\sqrt{2h^2}\right]\frac{\lambda}{2\sqrt{2}\varepsilon_0}\\
&=\left[h\ln\left(\frac{1}{\sqrt{2}-1}\right)+h-\sqrt{2}h\right]\frac{\lambda}{2\sqrt{2}\varepsilon_0}\\
&=\left[1-\sqrt{2}-\ln\left(\sqrt{2}-1\right)\right]\frac{\lambda h}{2\sqrt{2}\varepsilon_0}
\end{aligned}$$

2.2.8 一个半径为 r 的圆柱均匀带电, 电荷体密度为 ρ, 求圆柱中心上方距 h 处的电势

$$\begin{aligned}
V&=\int_{-h}^0 \frac{\rho}{2\varepsilon_0}\frac{r^2}{h+x_3+\sqrt{(h+x_3)^2+r^2}}\,\mathrm{d}x_3\\
&\xlongequal{u=h+x_3}\int_0^h \frac{\rho}{2\varepsilon_0}\frac{r^2}{u+\sqrt{u^2+r^2}}\,\mathrm{d}u\\
&=\int_0^h \frac{\rho r^2}{2\varepsilon_0}\frac{\sqrt{u^2+r^2}-u}{r^2}\,\mathrm{d}u\\
&=\frac{\rho}{2\varepsilon_0}\int_0^h \sqrt{u^2+r^2}-u\,\mathrm{d}u\\
&=u\sqrt{u^2+r^2}+r^2\ln\left(u+\sqrt{u^2+r^2}\right)-u^2\bigg|_0^h\frac{\rho}{4\varepsilon_0}\\
&=\left[h\sqrt{h^2+r^2}+r^2\ln\left(h+\sqrt{h^2+r^2}\right)-h^2-r^2\ln r\right]\frac{\rho}{4\varepsilon_0}
\end{aligned}$$

2.3 静电场的能量

2.3.1 离散电荷体系的静电能

考虑由 N 个点电荷 q_i 构成的静电体系。将电荷从无穷远逐个缓慢搬运到其最终位置, 外力所做的总功即为体系的静电能。

设在放置第 i 个电荷时, 其余电荷已就位, 则该电荷所处位置的当前电势为

$$\varphi_i = \sum_{j=1}^{i-1} \frac{1}{4\pi\varepsilon_0} \frac{q_j}{|\boldsymbol{r}_i - \boldsymbol{r}_j|}.$$

因此, 体系的总静电能为

$$U = \sum_{i=1}^{N} \sum_{j=1}^{i-1} \frac{1}{4\pi\varepsilon_0} \frac{q_i q_j}{|\boldsymbol{r}_i - \boldsymbol{r}_j|}.$$

注意到相互作用能在指标交换下满足 $U_{ij} = U_{ji}$, 而对所有 $i \neq j$ 的求和中每一对指标被计数两次, 故可将上式对称化为

$$U = \frac{1}{2} \sum_{\substack{i,j=1 \\ i \neq j}}^{N} \frac{1}{4\pi\varepsilon_0} \frac{q_i q_j}{|\boldsymbol{r}_i - \boldsymbol{r}_j|} = \frac{1}{2} q_i \varphi_i.$$

2.3.2 连续电荷分布的能量表达式

对离散情形的自然推广给出

$$U = \frac{1}{2} \iiint \rho\varphi \, \mathrm{d}V.$$

2.3.3 场能量

静电势满足泊松方程

$$\nabla^2 \varphi = -\frac{\rho}{\varepsilon_0}.$$

将其代入能量表达式, 得

$$U = -\frac{\varepsilon_0}{2} \iiint \varphi \, \nabla^2 \varphi \, \mathrm{d}V.$$

对右端积分作分部积分。注意到

$$\nabla \cdot (\varphi \nabla \varphi) = (\nabla \varphi)^2 + \varphi \nabla^2 \varphi,$$

于是

$$\varphi \nabla^2 \varphi = \nabla \cdot (\varphi \nabla \varphi) - (\nabla \varphi)^2.$$

代入得

$$U = -\frac{\varepsilon_0}{2} \left[\iiint \nabla \cdot (\varphi \nabla \varphi) \, \mathrm{d}V - \iiint (\nabla \varphi)^2 \, \mathrm{d}V \right].$$

当 $r \to \infty$ 时,

$$\varphi(\boldsymbol{r}) \sim \frac{1}{r}, \qquad \nabla \varphi \sim \frac{1}{r^2}.$$

因此

$$\varphi \nabla \varphi \sim \frac{1}{r^3},$$

对应的无穷远处曲面积分

$$\oiint \varphi \nabla \varphi \cdot \mathrm{d}\boldsymbol{S} \to 0$$

$$U = \frac{\varepsilon_0}{2} \iiint (\nabla \varphi)^2 \, \mathrm{d}V.$$

$$U = \frac{\varepsilon_0}{2} \iiint E^2 \, \mathrm{d}V$$

2.3.4 电场能量密度

这表明：静电能可以视为分布在空间中的电场所携带的能量。

由此自然引入电场的能量密度

$$\boxed{u \equiv \frac{\varepsilon_0}{2}\boldsymbol{E}^2}$$

2.3.5 考虑两个同心球面, 半径分别为 a 和 b, 内球面带有电荷 q , 外球面带有电荷 $-q$, 求总能量

$$U = \frac{\varepsilon_0}{2}\iiint E^2\,\mathrm{d}V = \frac{\varepsilon_0 4\pi}{2}\int_a^b \left(\frac{q}{4\pi\varepsilon_0 r^2}\right)^2 r^2\,\mathrm{d}r = \int_a^b \frac{q^2}{8\pi\varepsilon_0 r^2}\,\mathrm{d}r = \frac{q^2}{8\pi\varepsilon_0}\left(\frac{1}{b}-\frac{1}{a}\right)$$

2.4 导体

2.4.1 导体内部电场为零

若导体内部存在非零电场, 则自由电荷将在电场作用下持续运动, 与静电平衡的假设矛盾。因此, 导体内部必须满足

$$\boldsymbol{E} = -\boldsymbol{\nabla}\varphi = \boldsymbol{0}.$$

2.4.2 导体内部体电荷密度为零

$$\boldsymbol{\nabla}\cdot\boldsymbol{E} = \frac{\rho}{\varepsilon_0},$$

结合 $\boldsymbol{E} = \boldsymbol{0}$,

$$\rho = 0 \qquad (\text{导体内部}).$$

2.4.3 净电荷只能分布在导体表面

既然导体内部体电荷密度为零, 而导体整体可能带有净电荷, 则这些电荷只能分布在导体的表面上。

2.4.4 导体是等势体

由 $\boldsymbol{E} = -\boldsymbol{\nabla}\varphi$ 以及导体内部 $\boldsymbol{E} = \boldsymbol{0}$, 可知导体内部电势处处相同

2.4.5 导体表面外侧电场垂直于表面

考虑导体表面上一点的切向电场分量。若存在非零切向分量, 则自由电荷将在表面沿切向运动, 从而破坏静电平衡。

因此, 导体表面外侧的电场只能沿法向：

$$\boldsymbol{E}_{\parallel} = 0$$

2.4.6 空腔中含点电荷的球形导体

设一不带电的球形导体, 半径为 R, 中心位于原点, 其内部挖去一任意形状的空腔。在空腔内某处放置一点电荷 q。求球外区域的电场分布。

导体处于静电平衡时, 金属内部电场为零, 因而导体整体为等势体。取一紧贴导体内壁、位于金属内部的高斯面, 由高斯定律可得

$$\oint \boldsymbol{E} \cdot \mathrm{d}\boldsymbol{a} = 0,$$

故内壁所感应的总电荷必为 $-q$。又由于导体整体不带电, 外表面所带电荷总量为 $+q$。

关键在于确定外表面电荷的分布形式。注意到：在所有满足

$$\boldsymbol{E} = 0 \quad (\text{导体内部}), \qquad \varphi = \text{常数} \quad (\text{导体表面})$$

的允许电荷分布中, 实际的静电平衡态对应于体系总静电能的极小值。

球外区域不含自由电荷, 其电势完全由外表面电荷分布决定。若外表面电荷分布破坏球对称性, 则球外电场中将出现非径向分量, 从而在保持总电荷为 q 的约束下增加电场能

$$U = \frac{\varepsilon_0}{2} \int E^2 \, \mathrm{d}V.$$

因此, 能量极小所对应的电荷分布必然保持球对称性。

由此可知, 外表面电荷在球面上均匀分布, 其产生的球外电场与位于球心的点电荷 q 完全相同。因此, 球外任意一点处的电场为

$$\boldsymbol{E}(\boldsymbol{r}) = \frac{1}{4\pi\varepsilon_0} \frac{q}{r^2} \boldsymbol{e}_r, \qquad r > R.$$

可见, 空腔的形状以及点电荷在腔内的具体位置, 均不影响球外区域的电场分布。

2.4.7 一个半径为 R 的金属球, 带有电荷 q, 这个金属球又被一个厚的同心金属球壳所包围 (球壳内径为 a, 外径为 b)。

(a) 分别求出 R, a, b 球面上的电荷面密度 σ。

(b) 求出球心处的电势, 选无限远处为参考点。

(c) 现在球壳的外表面接地, 电势能为零。(a) 和 (b) 所得结果改变为什么?

(a):

$$\sigma_R = \frac{q}{4\pi R^2}$$
$$\sigma_a = \frac{q}{4\pi a^2}$$
$$\sigma_b = \frac{q}{4\pi b^2}$$

(b):

$$V = \int_b^\infty \frac{q}{4\pi\varepsilon_0 r} \, \mathrm{d}r + \int_R^a \frac{q}{4\pi\varepsilon_0 r} \, \mathrm{d}r = \frac{q}{4\pi\varepsilon_0} \ln\left(\frac{bR}{a}\right)$$

(c):

$$\begin{aligned}\sigma_R &= \frac{q}{4\pi R^2}\\ \sigma_a &= \frac{q}{4\pi a^2}\\ \sigma_b &= 0\\ V &= \frac{q}{4\pi\varepsilon_0}\ln\left(\frac{R}{a}\right)\end{aligned}$$

2.4.8 一个半径为 R 的导体球体, 其内部有两个半径分别为 a 和 b 的圆形空洞, 在 a 空洞的中心放有点电荷 q, 在 b 空洞的中心放有点电荷 q。
(a) 求出电荷面密度 σ_a, σ_b 和 σ_R。
(b) 导体外面的电场是什么?
(c) 每个空洞内的电场是什么?
(d)q_a 和 q_b 受到的力是什么?
(e) 如果让第三个电荷 q 靠近导体, 上面所得结果哪一个会发生变化?

(a):

$$\begin{aligned}\sigma_R &= \frac{2q}{4\pi R^2}\\ \sigma_a &= \frac{q}{4\pi a^2}\\ \sigma_b &= \frac{q}{4\pi b^2}\end{aligned}$$

(b):

$$\boldsymbol{E} = \frac{2q\boldsymbol{r}}{4\pi\varepsilon_0 r^3}$$

(c):

$$\begin{aligned}\boldsymbol{E_a} &= \frac{q\boldsymbol{r}}{4\pi\varepsilon_0 r^3}\\ \boldsymbol{E_b} &= \frac{q\boldsymbol{r}}{4\pi\varepsilon_0 r^3}\end{aligned}$$

(d):0

(e): σ_R, b

2.5 电容

$$C \equiv \frac{Q}{V} \tag{2.6}$$

$$W = \int_0^Q \left(\frac{q}{C}\right)\mathrm{d}q = \frac{Q^2}{2C} \tag{2.7}$$

2.5.1 两个同轴金属管壳, 半径分别为 a 和 b, 求出单位长度的电容。

$$
\begin{aligned}
V &= \int_a^b \frac{Q}{2\varepsilon_0 r}\,\mathrm{d}r \\
&= \frac{Q}{2\varepsilon_0}\ln\left(\frac{b}{a}\right) \\
C &= \frac{Q}{V} \\
&= \frac{2\varepsilon_0}{\ln\left(\frac{b}{a}\right)}
\end{aligned}
$$

2.6 习题

2.6.1 一个均匀带电、边长为 $2a$ 的正方形面, 电荷面密度为 σ。求出距中心高度为 z 处的电场。

$$
\begin{aligned}
&\int_0^a \frac{\sigma}{\pi\varepsilon_0\sqrt{2L^2+z^2}}\boldsymbol{e}_3\frac{2Lz}{L^2+z^2}\,\mathrm{d}L \\
\xlongequal{2L^2=z^2\tan^2\theta} &\int_0^{\arctan\frac{a}{\sqrt{2}z}} \frac{\sigma}{\pi\varepsilon_0\sqrt{z^2\tan^2\theta+z^2}}\boldsymbol{e}_3\frac{2z\tan\theta z}{z^2\tan^2\theta+2z^2}\,\mathrm{d}z\tan\theta \\
=&\int_0^{\arctan\frac{\sqrt{2}a}{z}} \frac{\sigma\cos\theta}{\pi\varepsilon_0 z\cos^2\theta}\boldsymbol{e}_3\frac{2z\tan\theta z}{z^2\tan^2\theta+2z^2}\,\mathrm{d}z\theta \\
=&\int_0^{\arctan\frac{\sqrt{2}a}{z}} \frac{\sigma}{\pi\varepsilon_0\cos\theta}\boldsymbol{e}_3\frac{2\tan\theta}{\tan^2\theta+2}\,\mathrm{d}\theta \\
=&\int_0^{\arctan\frac{\sqrt{2}a}{z}} \frac{2\sigma}{\pi\varepsilon_0}\boldsymbol{e}_3\frac{\tan\theta}{\tan^2\theta\cos\theta+2\cos\theta}\,\mathrm{d}\theta \\
=&\int_0^{\arctan\frac{\sqrt{2}a}{z}} \frac{2\sigma}{\pi\varepsilon_0}\boldsymbol{e}_3\frac{\sin\theta}{\sin^2\theta+2\cos^2\theta}\,\mathrm{d}\theta \\
=&-\int_0^{\arctan\frac{\sqrt{2}a}{z}} \frac{2\sigma}{\pi\varepsilon_0}\boldsymbol{e}_3\frac{1}{1+\cos^2\theta}\,\mathrm{d}\cos\theta \\
=&-\frac{2\sigma}{\pi\varepsilon_0}\boldsymbol{e}_3\arctan\cos\theta\Big|_0^{\arctan\frac{\sqrt{2}a}{z}} \\
=&\frac{2\sigma}{\pi\varepsilon_0}\boldsymbol{e}_3\left(\frac{\pi}{4}-\arctan\frac{z}{\sqrt{2a^2+z^2}}\right)
\end{aligned}
$$

2.6.2 已知电场 $\boldsymbol{E}=\dfrac{A\boldsymbol{e}_r+B\sin\theta\cos\phi\boldsymbol{e}_\phi}{r}$, 求电荷密度

$$
\begin{aligned}
\boldsymbol{\nabla}\cdot\boldsymbol{E}==&\frac{1}{sr}\frac{\partial srE_r}{\partial r}+\frac{1}{s}\frac{\partial E_\phi}{\partial\phi}\\
=&\frac{1}{sr}\frac{\partial sA}{\partial r}+\frac{1}{sr}\frac{\partial B\sin\theta\cos\phi}{\partial\phi}\\
=&\frac{A}{r^2}-\frac{1}{r^2}B\sin\phi
\end{aligned}
$$

2.6.3 一个均匀带电球体, 求出南半球与北半球之间的净相互作用力

$$
\begin{aligned}
E_z=E_r\cos\theta=&\frac{\rho\cos\theta}{4\pi\varepsilon_0r^2}\frac{4\pi r^3}{3}=\frac{\rho\cos\theta r}{3\varepsilon_0}\\
F=&\int_0^R\mathrm{d}r\int_0^{2\pi}\mathrm{d}\phi\int_0^{\frac{\pi}{2}}r^2\sin\theta\rho E_z\,\mathrm{d}\theta\\
=&\int_0^R\mathrm{d}r\int_0^{2\pi}\mathrm{d}\phi\int_0^{\frac{\pi}{2}}r^3\sin\theta\rho\frac{\rho\cos\theta}{3\varepsilon_0}\,\mathrm{d}\theta\\
=&\pi\int_0^R r^3\,\mathrm{d}r\int_0^{\frac{\pi}{2}}\sin\theta\rho^2\frac{2\cos\theta}{3\varepsilon_0}\,\mathrm{d}\theta\\
=&\pi R^4\int_0^{\frac{\pi}{2}}\sin\theta\rho^2\frac{\cos\theta}{6\varepsilon_0}\,\mathrm{d}\theta\\
=&\pi R^4\int_0^{\frac{\pi}{2}}\rho^2\frac{\sin2\theta}{24\varepsilon_0}\,\mathrm{d}2\theta\\
=&-\rho^2R^4\pi\frac{\cos2\theta}{24\varepsilon_0}\bigg|_0^{\frac{\pi}{2}}\\
=&\frac{\rho^2R^4\pi}{12\varepsilon_0}
\end{aligned}
$$

2.6.4 一个半径为 R 的倒置半球面均匀带电, 电荷面密度为 σ。求出北极与球心处的电势差

$$
\begin{aligned}
V_{\text{半球北极}} &= \mathrm{d}\varphi \int_0^{\frac{\pi}{2}} \frac{\sigma R \sin\theta}{2\varepsilon_0} \frac{1}{\sqrt{R^2\left(1-\cos\theta\right)^2 + R^2\sin^2\theta}} R\,\mathrm{d}\theta \\
&= \int_0^{\frac{\pi}{2}} \frac{\sigma \sin^2\theta}{2\varepsilon_0} \frac{1}{\sqrt{\left(1-\cos\theta\right)^2 + \sin^2\theta}} R^2\,\mathrm{d}\theta \\
&= \int_0^{\frac{\pi}{2}} \frac{\sigma \sin\theta}{2\varepsilon_0} \frac{1}{\sqrt{2-2\cos\theta}} R\,\mathrm{d}\theta \\
&= \int_0^{\frac{\pi}{2}} \frac{\sigma 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{2\sqrt{2}\varepsilon_0} \frac{1}{\sqrt{1-\cos^2\frac{\theta}{2}+\sin^2\frac{\theta}{2}}} R\,\mathrm{d}\theta \\
&= \int_0^{\frac{\pi}{2}} \frac{\sigma \sin\frac{\theta}{2}\cos\frac{\theta}{2}}{\sqrt{2}\varepsilon_0} \frac{1}{\sqrt{2\sin^2\frac{\theta}{2}}} R\,\mathrm{d}\theta \\
&= \int_0^{\frac{\pi}{2}} \frac{\sigma \cos\frac{\theta}{2}}{2\varepsilon_0} R\,\mathrm{d}\theta \\
&= \int_0^{\frac{\pi}{2}} \frac{\sigma}{\varepsilon_0} R\,\mathrm{d}\sin\frac{\theta}{2} \\
&= \left.\frac{\sigma \sin\frac{\theta}{2}}{\varepsilon_0} R\right|_0^{\frac{\pi}{2}} \\
&= \frac{\sqrt{2}\sigma}{2\varepsilon_0} R \\
V_{\text{半球球心}} &= \frac{2\pi R^2 \sigma}{4\pi\varepsilon_0 R} = \frac{\sigma R}{2\varepsilon_0} \\
\Delta V &= \frac{\sigma}{2\varepsilon_0}\frac{R^2}{\sqrt{R^2}} + \frac{\sigma R}{2\varepsilon_0} - \frac{\sigma}{2\varepsilon_0}\frac{R^2}{R+\sqrt{R^2+R^2}} - \frac{\sqrt{2}\sigma}{2\varepsilon_0} R \\
&= \frac{\sigma R}{2\varepsilon_0} - \frac{\sqrt{2}\sigma}{2\varepsilon_0} R \\
&= \frac{\sigma R}{2\varepsilon_0} - \frac{\sqrt{2}\sigma}{2\varepsilon_0} R
\end{aligned}
$$

2.6.5 一个半径为 R 的球体, 电荷密度 $\rho = kr$, 求能量

$$
\begin{aligned}
|\boldsymbol{E}| &= \begin{cases} \dfrac{1}{4\pi\varepsilon_0 x_3^2}\displaystyle\int_0^R 4\pi k r^3\,\mathrm{d}r\,(x_3 \geq R) \\ \dfrac{1}{4\pi\varepsilon_0 x_3^2}\displaystyle\int_0^{x_3} 4\pi k r^3\,\mathrm{d}r\,(x_3 < R) \end{cases} \\
&= \begin{cases} \dfrac{1}{4\varepsilon_0 x_3^2} k R^4\,(x_3 \geq R) \\ \dfrac{1}{4\varepsilon_0 x_3^2} k x_3^4\,(x_3 < R) \end{cases} \\
&= \begin{cases} \dfrac{kR^4}{4\varepsilon_0 x_3^2}\,(x_3 \geq R) \\ \dfrac{k x_3^2}{4\varepsilon_0}\,(x_3 < R) \end{cases} \\
U &= \frac{\varepsilon_0}{2} 4\pi \int_0^R \left(\frac{k x_3^2}{4\varepsilon_0}\right)^2 x_3^2\,\mathrm{d}x_3 + \frac{\varepsilon_0}{2} 4\pi \int_R^\infty \left(\frac{kR^4}{4\varepsilon_0 x_3^2}\right)^2 x_3^2\,\mathrm{d}x_3 \\
&= 2\pi \int_0^R \frac{k^2 x_3^6}{16\varepsilon_0}\,\mathrm{d}x_3 + 2\pi \int_R^\infty \frac{k^2 R^8}{16\varepsilon_0 x_3^2}\,\mathrm{d}x_3 \\
&= \pi \frac{k^2 R^7}{56\varepsilon_0} + \pi \frac{k^2 R^8}{8\varepsilon_0 R} \\
&= \pi \frac{k^2 R^7}{7\varepsilon_0}
\end{aligned}
$$

2.6.6 电势为 $V=\frac{Ae^{-\lambda r}}{r}$, 求电场, 电荷密度, 总电荷

$$
\begin{aligned}
\boldsymbol{E} &= -\boldsymbol{\nabla} V \\
&= \frac{\partial \dfrac{Ae^{-\lambda r}}{r}}{\partial r}\boldsymbol{e}_r \\
&= \frac{\lambda r A e^{-\lambda r} + Ae^{-\lambda r}}{r^2}\boldsymbol{e}_r \\
\rho &= \varepsilon_0 \boldsymbol{\nabla}^2 V \\
&= \varepsilon_0 \frac{1}{r^2}\frac{\partial r^2 a_r}{\partial r} \\
&= \varepsilon_0 \frac{1}{r^2}\frac{\partial \lambda r A e^{-\lambda r} + Ae^{-\lambda r}}{\partial r} \\
&= \varepsilon_0 \frac{\lambda A e^{-\lambda r} - \lambda^2 r A e^{-\lambda r} - \lambda A e^{-\lambda r}}{r^2} \\
&= -\frac{\varepsilon_0 \lambda^2 A e^{-\lambda r}}{r}\ (r \neq 0) \\
\rho &= \frac{\varepsilon_0 A 4\pi \delta^{(3)}(r)}{r} - \frac{\varepsilon_0 \lambda^2 A e^{-\lambda r}}{r} \\
Q &= 4\pi \int_0^\infty \rho r^2 \,\mathrm{d}r \\
&= 0
\end{aligned}
$$

2.6.7 两条平行于 z 轴的无限长均匀带电线, 电荷线密度分别为 $+\lambda$ 和 $-\lambda$, 距离为 $2d$。
(a) 求出任意一点的电势。
(b) 证明等势面为圆柱面, 对给定的电势 V, 给出圆柱面的半径和轴的位置。

(a):

$$
\begin{aligned}
V_{+\lambda} &= -\frac{\lambda}{2\pi\varepsilon_0}\ln|\boldsymbol{r} - d\boldsymbol{e}_1| \\
V_{-\lambda} &= \frac{\lambda}{2\pi\varepsilon_0}\ln|\boldsymbol{r} + d\boldsymbol{e}_1| \\
V &= -\frac{\lambda}{2\pi\varepsilon_0}\ln|\boldsymbol{r} - d\boldsymbol{e}_1| + \frac{\lambda}{2\pi\varepsilon_0}\ln|\boldsymbol{r} + d\boldsymbol{e}_1| \\
&= \frac{\lambda}{4\pi\varepsilon_0}\ln\left[\frac{(x_1+d)^2 + x_2^2}{(x_1-d)^2 + x_2^2}\right]
\end{aligned}
$$

(b):

$$
\begin{aligned}
(x_1+d)^2+x_2^2 &= k\left[(x_1-d)^2+x_2^2\right] \\
x_1^2+2x_1d+d^2+x_2^2 &= kx_1^2-2kx_1d+kd^2+kx_2^2 \\
0 &= (k-1)x_1^2-2(k+1)x_1d+(k-1)d^2+(k-1)x_2^2 \\
0 &\xlongequal{u=\frac{k+1}{k-1}} x_1^2-2ux_1d+d^2+x_2^2 \\
u^2-d^2 &= (x_1-u)^2+x_2^2 \\
R &= \sqrt{u^2-d^2} \\
&= \sqrt{\left(\frac{k+1}{k-1}\right)^2-d^2} \\
&= \sqrt{\left(\frac{e^{\frac{4\pi\varepsilon_0 V}{\lambda}}+1}{e^{\frac{4\pi\varepsilon_0 V}{\lambda}}-1}\right)^2-d^2}
\end{aligned}
$$

2.6.8 在一个真空二极管中, 电子从阴极面“热蒸发”后向阳极面加速运动, 阴极电势为零, 对面阳极的电势为 V_0。在两极间隙中所形成的电子云 (称为空间电荷) 很快会达到一种分布状态, 使得阴极面上的电场为零。然后在两极板之间形成稳定的电流 **I**。假定两个极板面积 A 远大于它们之间的距离 $d(A>>d)$, 所以边界效应可以忽略。则 V,ρ,v(电子速度) 都仅是 x 的函数。

(a) 写出在两极板之间空间的泊松方程。

(b) 假定电子从阴极是从静止开始运动的, 那么在点 x, 这里电势为 $V(x)$, 电子速度为多少?

(c) 在稳定状态下, 电流 I 不依赖于 x。那么 ρ 和 v 之间的关系是什么?

(d) 利用上面的结果, 消去 ρ 和 v, 得出 V 满足的微分方程。

(e) 作为 x,V_0,d 的函数, 求出 V 的解。并与没有空间电荷的情况比较。另外作为 x 的函数, 求出 ρ 和 v。

(f) 证明 $1=KV_0^{\frac{3}{2}}$ 求出常数 K。

(a):

$$
\frac{\partial^2 V}{\partial x^2}=\frac{\rho}{\varepsilon_0}
$$

(b):

$$
\begin{aligned}
eV &= \frac{1}{2}mv^2 \\
v &= \sqrt{\frac{2eV}{m}}
\end{aligned}
$$

(c):

$$
\begin{aligned}
I &= \frac{\Delta q}{\Delta t} \\
&= \rho A v
\end{aligned}
$$

(d):

$$\begin{aligned}
\frac{\partial^2 V}{\partial x^2} &= \frac{\rho}{\varepsilon_0} \\
\frac{\partial^2 V}{\partial x^2} &= \frac{I}{Av\varepsilon_0} \\
\frac{\partial^2 V}{\partial x^2} &= \frac{I}{A\sqrt{\frac{2eV}{m}}\varepsilon_0}
\end{aligned}$$

(e):

$$\begin{aligned}
2\frac{\partial V}{\partial x}\frac{\partial^2 V}{\partial x^2} &= 2\frac{\partial V}{\partial x}\frac{I}{A\sqrt{\frac{2eV}{m}}\varepsilon_0} \\
\frac{\partial}{\partial x}\left(\frac{\partial V}{\partial x}\right)^2 &= 4\frac{\partial V^{\frac{1}{2}}}{\partial x}\frac{I}{A\sqrt{\frac{2e}{m}}\varepsilon_0} \\
\left(\frac{\partial V}{\partial x}\right)^2 &= 4\frac{IV^{\frac{1}{2}}}{A\sqrt{\frac{2e}{m}}\varepsilon_0} \\
\frac{\partial V}{\partial x} &= 2\frac{I^{\frac{1}{2}}V^{\frac{1}{4}}}{A^{\frac{1}{2}}\left(\frac{2e}{m}\right)^{\frac{1}{4}}\varepsilon_0^{\frac{1}{2}}} \\
V^{-\frac{1}{4}}\partial V &= 2\frac{I^{\frac{1}{2}}m^{\frac{1}{4}}\partial x}{A^{\frac{1}{2}}(2e)^{\frac{1}{4}}\varepsilon_0^{\frac{1}{2}}} \\
\frac{4}{3}V^{\frac{3}{4}} &= 2\frac{I^{\frac{1}{2}}m^{\frac{1}{4}}x}{A^{\frac{1}{2}}(2e)^{\frac{1}{4}}\varepsilon_0^{\frac{1}{2}}} \\
V &= \frac{3^{\frac{4}{3}}I^{\frac{2}{3}}x^{\frac{4}{3}}m^{\frac{1}{4}}}{2A^{\frac{1}{2}}e^{\frac{1}{3}}\varepsilon_0^{\frac{2}{3}}} \\
v &= \sqrt{\frac{2eV}{m}} \\
&= \sqrt{\frac{2e\frac{3^{\frac{4}{3}}I^{\frac{2}{3}}x^{\frac{4}{3}}m^{\frac{1}{4}}}{2A^{\frac{1}{2}}e^{\frac{1}{3}}\varepsilon_0^{\frac{2}{3}}}}{m}} \\
&= \sqrt{\frac{3^{\frac{4}{3}}I^{\frac{2}{3}}x^{\frac{4}{3}}e^{\frac{2}{3}}}{A^{\frac{1}{2}}\varepsilon_0^{\frac{2}{3}}m^{\frac{3}{4}}}} \\
&= \frac{3^{\frac{2}{3}}I^{\frac{1}{3}}x^{\frac{2}{3}}e^{\frac{1}{3}}}{A^{\frac{1}{4}}\varepsilon_0^{\frac{1}{3}}m^{\frac{3}{8}}} \\
\rho &= \varepsilon_0\frac{\partial^2 V}{\partial x^2} \\
&= \frac{2I^{\frac{2}{3}}m^{\frac{1}{4}}\varepsilon_0^{\frac{1}{3}}}{3^{\frac{2}{3}}A^{\frac{1}{2}}e^{\frac{1}{3}}x^{\frac{2}{3}}}
\end{aligned}$$

(f):

$$
\begin{aligned}
V &= \frac{3^{\frac{4}{3}} I^{\frac{2}{3}} x^{\frac{4}{3}} m^{\frac{1}{4}}}{2A^{\frac{1}{2}} e^{\frac{1}{3}} \varepsilon_0^{\frac{2}{3}}} \\
I^{\frac{2}{3}} &= \frac{V 2A^{\frac{1}{2}} e^{\frac{1}{3}} \varepsilon_0^{\frac{2}{3}}}{3^{\frac{4}{3}} x^{\frac{4}{3}} m^{\frac{1}{4}}} \\
I &= \frac{V^{\frac{3}{2}} 2^{\frac{3}{2}} A^{\frac{3}{4}} e^{\frac{1}{2}} \varepsilon_0^{\frac{2}{3}}}{3^2 x^2 m^{\frac{3}{2}}} \\
I &= \frac{V_0^{\frac{3}{2}} 2^{\frac{3}{2}} A^{\frac{3}{4}} e^{\frac{1}{2}} \varepsilon_0^{\frac{2}{3}}}{3^2 d^2 m^{\frac{3}{2}}}
\end{aligned}
$$

2.6.9 假设现在极精确的测量已经揭示出库仑定律的误差。两个点电荷之间的作用力为 $\boldsymbol{F} = \frac{q_1 q_2}{4\pi\varepsilon_0 r^2}\left(1+\frac{r}{\lambda}\right) e^{-\frac{r}{\lambda}} \boldsymbol{e}_r$ 式中，λ 是一个新的自然常数。你的任务是按照这个新发现重新表述静电学。假定叠加原理仍然成立。

(a) 电场是什么?

(b) 求出一个点电荷的电势。

(c) 对一个位于原点的点电荷, 证明 $\oiint_S \boldsymbol{E}\cdot \mathrm{d}\boldsymbol{S} + \frac{1}{\lambda^2}\iiint_V V\,\mathrm{d}\tau = \frac{q}{\varepsilon_0}$

(d) 证明 $\oiint_S \boldsymbol{E}\cdot \mathrm{d}\boldsymbol{S} + \frac{1}{\lambda^2}\iiint_V V\,\mathrm{d}\tau = \iiint_V \frac{\rho}{\varepsilon_0}\,\mathrm{d}\tau$

(a):

$$
\boldsymbol{E} = \frac{q}{4\pi\varepsilon_0 r^2}\left(1+\frac{r}{\lambda}\right) e^{-\frac{r}{\lambda}} \boldsymbol{e}_r
$$

(b):

$$
\begin{aligned}
V &= \int_R^\infty \frac{q}{4\pi\varepsilon_0 r^2}\left(1+\frac{r}{\lambda}\right) e^{-\frac{r}{\lambda}}\,\mathrm{d}r \\
&= \frac{q}{4\pi\varepsilon_0 \lambda}\int_R^\infty \left(\frac{\lambda^2}{r^2}+\frac{\lambda}{r}\right) e^{-\frac{r}{\lambda}}\,\mathrm{d}\frac{r}{\lambda} \\
&= -\frac{q}{4\pi\varepsilon_0 \lambda}\int_R^\infty \left(\frac{\lambda^2}{r^2}--\frac{\lambda}{r}\right) e^{-\frac{r}{\lambda}}\,\mathrm{d}-\frac{r}{\lambda} \\
&\xlongequal{u=-\frac{r}{\lambda}} -\frac{q}{4\pi\varepsilon_0 \lambda}\int_{-\frac{R}{\lambda}}^{-\infty} \left(\frac{1}{u^2}-\frac{1}{u}\right) e^{u}\,\mathrm{d}u \\
&= -\left.\frac{1}{u}e^{u}\right|_{-\frac{R}{\lambda}}^{-\infty} \frac{q}{4\pi\varepsilon_0 \lambda} \\
&= e^{-\frac{R}{\lambda}}\frac{q}{4\pi\varepsilon_0 R}
\end{aligned}
$$

(c):

$$
\begin{aligned}
\oiint_S \boldsymbol{E}\cdot \mathrm{d}\boldsymbol{S} &= 4\pi R^2 \frac{q}{4\pi\varepsilon_0 R^2}\left(1+\frac{R}{\lambda}\right)e^{-\frac{r}{\lambda}} = \frac{q}{\varepsilon_0}\left(1+\frac{R}{\lambda}\right)e^{-\frac{r}{\lambda}} \\
\frac{1}{\lambda^2}\iiint_V V\,\mathrm{d}\tau &= 4\pi\frac{1}{\lambda^2}\int_0^R e^{-\frac{r}{\lambda}}\frac{q}{4\pi\varepsilon_0 r}r^2\,\mathrm{d}r \\
&= \int_0^R -\frac{r}{\lambda}e^{-\frac{r}{\lambda}}\frac{q}{\varepsilon_0}\,\mathrm{d}-\frac{r}{\lambda} \\
&\xlongequal{u=-\frac{r}{\lambda}} \int_0^{-\frac{R}{\lambda}} ue^u\frac{q}{\varepsilon_0}\,\mathrm{d}u \\
&= (u-1)\,e^u\frac{q}{\varepsilon_0}\bigg|_0^{-\frac{R}{\lambda}} \\
&= \left(-\frac{R}{\lambda}-1\right)e^{-\frac{R}{\lambda}}\frac{q}{\varepsilon_0}+\frac{q}{\varepsilon_0} \\
\frac{q}{\varepsilon_0} &= \oiint_S \boldsymbol{E}\cdot\mathrm{d}\boldsymbol{S}+\frac{1}{\lambda^2}\iiint_V V\,\mathrm{d}\tau
\end{aligned}
$$

(d):